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TJA1104 application note

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Application note
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Document information

Information	Content
Keywords	Automotive Ethernet, IEEE 100BASE-T1, PHY, TJA1104A, TJA1104B, ASIL-B, MACsec
Abstract	The TJA1104 is a 100BASE-T1 compliant Ethernet PHY with MACsec. The device provides 100Mbit/s transmit and receive capability over unshielded twisted-pair cables, supporting a cable length of up to 15 m. It is available in two variants. The TJA1104A/C offers standard MII/RMII and RGMII(-ID) MAC interfaces. The variant TJA1104B/D provides SGMII.

1 Introduction

The TJA1104 is a 100BASE-T1 compliant Ethernet PHY. It provides 100 Mbit/s transmit and receive capability over unshielded twisted-pair cables, supporting a cable length of up to 15 m. The TJA1104A/C variants include standard MII/RMII and RGMII(-ID) MAC interfaces. The TJA1104B/D variant features an SGMII interface.

A MACSEC engine providing encryption, decryption, signing as well as signature checking is embedded in the PHY, supporting these operations at line rate (100 Mbit/s).

For sustainable system efficiency and flexibility, the device was optimized for lowest power consumption and highest robustness, while retaining a small package footprint.

The TJA1104 was developed in compliance with ISO26262, achieving ASIL-B. Adequate safety features have been implemented on-chip to support functional safety concepts at a system level. Additional documentation, including a safety manual, is available on request. In combination with state-of-the-art diagnostic features, the device is perfectly suited to support high availability of communication paths in upcoming automotive electrical/electronic (E/E) architectures.

The TJA1104 is designed to meet the harsh requirements of automotive applications. It is typically used in electronic control units (ECUs) like gateways, telematics and head units, as well as typical end nodes like radar, LiDAR, and camera sensors for driver assistance systems (ADAS).

The device supports OPEN Alliance TC-10 sleep/wake over data-line. By eliminating the need for extra cabling, the feature saves on cost and weight.

Due to its efficient IEEE1588v2 and 802.1AS-2020 compliant time stamping capability, the TJA1104 provides high-resolution accuracy for global time synchronization, which is required to support the timing critical data fusion functions of self-driving cars.

For more information, please visit https://www.nxp.com/TJA1104 for the TJA1104 data sheet.

2 100BASE-T1 Basics

2.1 System diagram

The 100BASE-T1 compliant Ethernet PHY provides 100 Mbit/s transmit and receive capability over a single unshielded twisted-pair cable. The device supports a cable length of at least 15 m with a bit error rate less than or equal to 10^{-10}. It is optimized for capacitive signal coupling to the twisted-pair lines. A common mode choke (CMC) is typically inserted into the signal path to comply with stringent automotive EMC requirements. Standard MII, RMII, or RGMII interfaces (TJA1104A/C) or SGMII interface (TJA1104B/D) can be used to connect the PHY with a Media Access Control (MAC) unit.

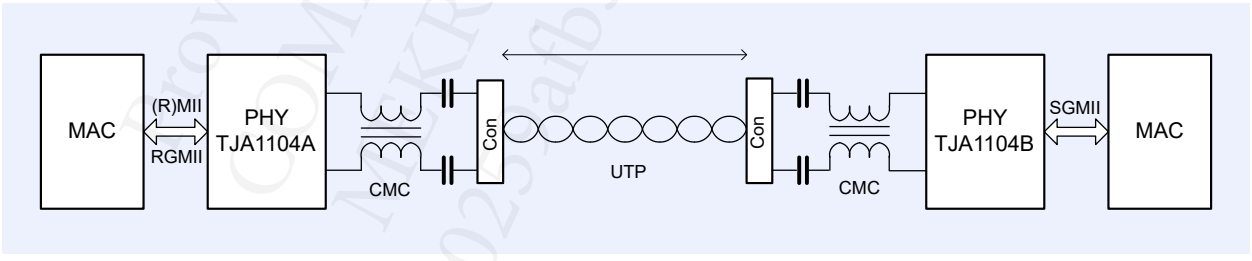


Figure 1. TJA1104A/B system diagram

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2.2 Link training

The PHY Control state machine (IEEE 802.3bw standard [3]) controls the link startup procedure. The state machine is shown in Figure 2. A 100BASE-T1 PHY is either in MASTER or SLAVE mode. To allow link-up, one device must be MASTER and the other SLAVE. This setting is controlled by pin strapping or via register control.

Before a link is established, a training phase (see TRAINING state) is needed. Training can last up to 100 ms, but is typically much faster.

The DEVICE_CONTROL.START_OPERATION bit (30.0040h.0) must be set before link training starts. This bit set automatically if the device is configured in AUTONOMOUS mode via pin strapping. Otherwise software must set this bit.

Once in TRAINING, the MASTER PHY initiates the training phase by sending an active idle sequence (tx_mode = SEND_I). The SLAVE PHY synchronizes its internal clock and locks the scrambler. Eventually the SLAVE PHY also goes into TRAINING state and sends SEND_I. In case one of the two PHYs is not yet ready (rem_rcvr_status = NOT_OK), the other PHY keeps sending the training pattern (SEND IDLE).

Once both PHYs agree that they are ready (loc_rcvr_status=OK and rem_rcvr_status=OK) they enter SEND IDLE OR DATA state. From that point onwards the bidirectional link is established and data communication is possible.

Since the link training can take up to 100 ms, the current link status should be checked.

LINK_AVAILABLE is defined in PHY_STATUS (30.8102h).

Table 1. Link status flags - PHY_STATUS (30.8102h)

Field	Bit	Description
LINK_AVAILABLE	6	Link is ready; if this bit is set, all other status flags are set
REM_RCVR_STATUS	4	Remote receiver status is set
SCR_STATUS	3	The scrambler is synchronized
LINK_STATUS	2	Link status from link monitor
LOC_RCVR_STATUS	1	Local receiver status is set

Alternatively, the LINK_STATUS_EVENT interrupt (PHY_FUNCTIONAL_IRQ_SOURCES.LINK_STATUS_EVENT - 30.80A0h.1) can be used to indicate an established link.

If no link is established within 100 ms, an optional cable test can be started. For details refer to Section 5.2

In case the TRAINING_FAIL interrupt is enabled and occurs it indicates that no link was established within 200 ms. The PHY automatically restarts training. There is no need to trigger a training restart manually. The LINK_TRAINING_TIMER (30.8340h.11:4) measures the training time. The register contains the time in milliseconds between entering SLAVE_SILENT and setting LINK_AVAILABILITY=1. If the PHY cycles through multiple training attempts, only the last, successful training is reflected in the register.

A SLAVE PHY can automatically detect and correct the cable polarity. This feature is enabled via pin strapping or register setting. See Section 4.7 for details.

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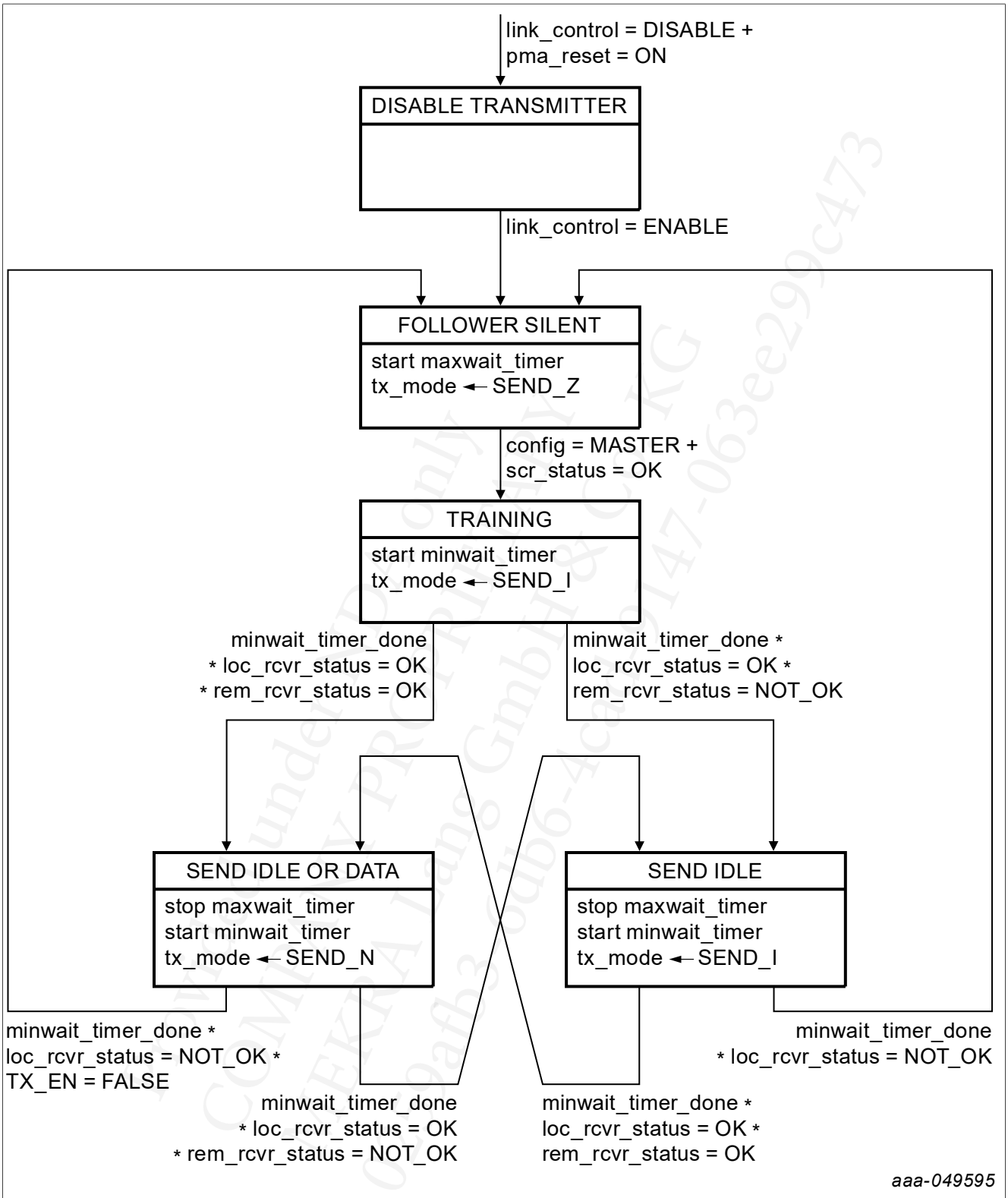


Figure 2. IEEE 802.3bw PHY control state machine

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3 Hardware Application

3.1 Overview

A typical application circuit for the TJA1104 can be seen in Figure 3 for the A variant and in Figure 4 for the B variant. Details of the MDI interface are shown in Section 3.4. The application shown contains capacitive coupling to the twisted-pair wires. To comply with EMC requirements a common-mode choke (CMC) should be inserted into the signal path. The coupling capacitors should be placed between common mode choke and connector. The MDI pins can withstand ESD events of at least ±6 kV according to IEC61000-4-2 and ±4 kV according to human-body model (HBM). In case the application demands exceed these limits, external ESD components are required. The device can withstand ± 500 V on all pins according to charge-device model (CDM).

A differential 100 Ω termination is integrated into the PHY. A common-mode termination network is recommended to avoid a floating common-mode behavior on the cable.

The TJA1104 has various power supply options. The device can be supplied from a single 3.3 V / 2.5 V source. Here, the internal LDOs generate the required 2.5 V and 1.1 V. It is also possible to supply all required voltages externally. Although the TJA1104 is already optimized for power consumption, this option is more efficient. The IO voltage can be selected between 1.8 V, 2.5 V, or 3.3 V. Details are given in Section 3.5

Control and status information is exchanged with the host controller via SMI interface. The INH control output of the PHY can optionally be used to switch off the main voltage regulator when the PHY is in Sleep mode. (See Section 6)

More detailed application information is provided in the following sections.

3.2 TJA1104A: MII/RMII/RGMII interface

The TJA1104A supports a number of xMII modes that can be selected via pin strapping or MDIO registers. The following list shows the available xMII modes and selectable submodes. For more details, refer to xMII interfaces section of the TJA1104 Datasheet [1].

Note: *xMII modes listed in Table 2 are only available in the TJA1104A variant.*

Table 2. xMII Modes

xMII mode	Short description
MII	Standard PHY MII mode
rev-MII	Reversed MII. The PHY mimics a MAC MII interface.
RMII	Standard RMII interface. REF_CLK is input for the device
RMII-clkmode	Standard RMII interface. REF_CLK is input for the device and used as a digital clock. In this mode, no external crystal is required
rev-RMII	Standard RMII interface. REF_CLK is output for the device
RGMII	RGMII 1.3 mode. Delays are expected on the PCB or the MAC side.
RGMII-ID (TXC)[1]	RGMII 2.0 mode. The TXC has internal delays enabled. The RXC has internal delays disabled.
RGMII-ID (TXC and RXC)	RGMII 2.0 mode. The TXC has internal delays enabled. The RXC has internal delays enabled.
RGMII-ID (RXC)	RGMII 2.0 mode. The TXC has internal delays disabled. The RXC has internal delays enabled.

[1] Not available via pin strapping

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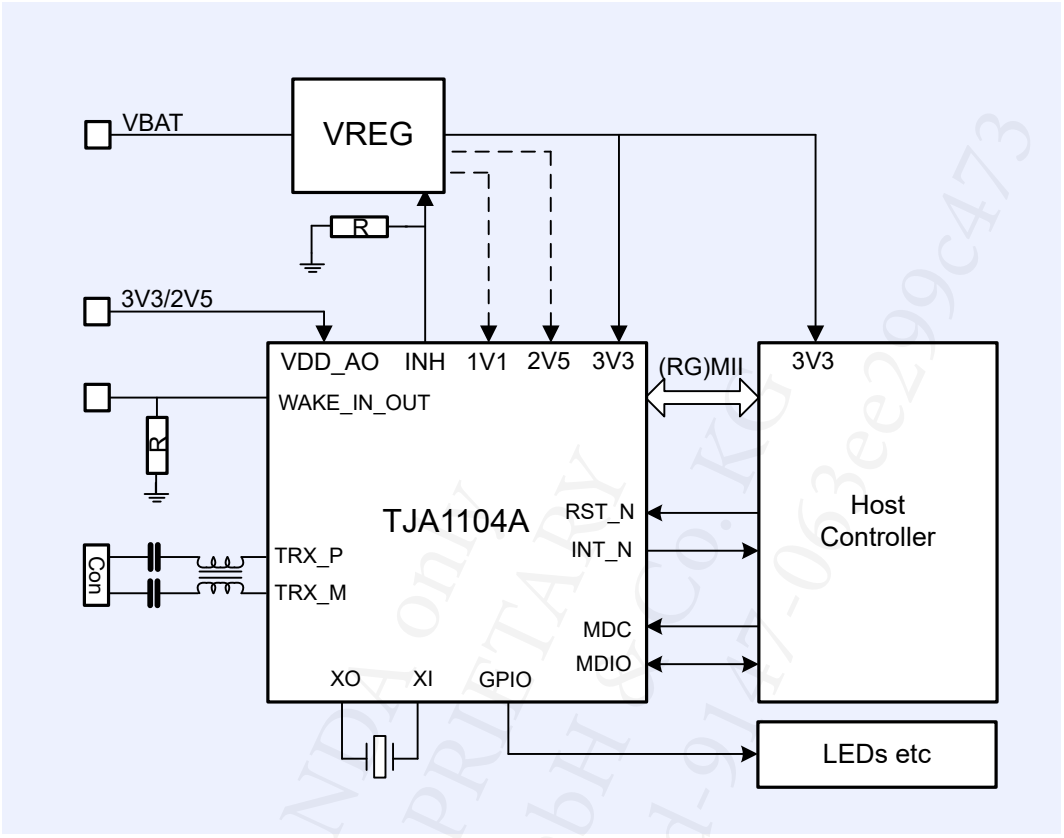


Figure 3. TJA1104A connection diagram

3.2.1 TX_ER and RX_ER

TX_ER and RX_ER are optional signals in MII/RMII modes. By default the pins are configured for TX_ER and RX_ER operation but can be disabled via register configuration and used as GPIOs.

RX_ER is a PHY output which is asserted to indicate invalid frame reception. TX_ER is a PHY input. When TX_ER is asserted, the PHY propagates the error indication and the remote MAC eventually discards the frame.

The role of the TX_ER pin is configured via register XMII_DATA_CONFIG.10 (30.AFCAh.10). The role of RX_ER is configured via XMII_DATA_CONFIG.2 (30.AFCAh.2).

3.2.2 xMII elastic buffers

The xMII interface uses elasticity buffers to compensate between clock differences of the 100BASE-T1 link and the xMII clocks. Essentially this allows using 100 ppm clock sources even in modes that require 50 ppm clock accuaries. Details can be found in the datasheet [1] under section 6.4.6. In general the following statement must be evaluated:

In xMII modes that require 50 ppm precision a 100 ppm clock can be used if either the MAC is only clock sender (RMII / RMII-clkmode) or MAC also allows 100 ppm precision (RGMII / RMII-rev)

The table below gives some practical guidelines.

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Table 3. Reset options

xMII mode	clock precision according to specification	clock precision respecting elastic buffers
MII	100 ppm	100 ppm[1]
MII-rev	100 ppm	100 ppm[1]
RMII	50 ppm	100 ppm
RMII-rev	50 ppm	50 ppm 100 ppm[2]
RMII-clkmode	50 ppm	100 ppm[3]
RGMII	50 ppm	50 ppm 100 ppm[4]

[1] Elastic buffers not relevant as specification only requires 100 ppm.
[2] Only if MAC allows 100 ppm xMII precision - check with MAC supplier.
[3] The PHY is entirely clocked by the RMII clock. Therefore there is no deviation between RMII and PHY clock. 100 ppm are required by 100BASE-T1 specification.
[4] Only if MAC allows 100 ppm xMII precision - check with MAC supplier

3.2.3 xMII isolation

The xMII data and control signals can be isolated, meaning disconnected between internal logic and IO-cells. There are several ways of achieving this:

- Logical isolate (RX+TX): configurable via MMD30.XMII_CONTROL[LOGICAL_ISOLATE] (30.AFC0h.2) - TX data and control signals are ignored, RX data and control signals are tied LOW. Output clocks remain active.
- RXD disable (RX only): configurable via MMD30.XMII_DATA_CONFIG[RXD_DISABLE] - RX data and control signals are tied LOW. Output clocks remain active.

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3.3 TJA1104B/D: SGMII interface

The TJA1104B/D uses a Serial-GMII (SGMII) SerDes interface for the PHY-MAC or PHY-PHY connection. Compared to RGMII, SGMII uses fewer interface signals. EMC performance is different due to smaller voltage swings and changed coding scheme compared to RGMII.

Note: *SGMII is only available in the TJA1104B/D variants.*

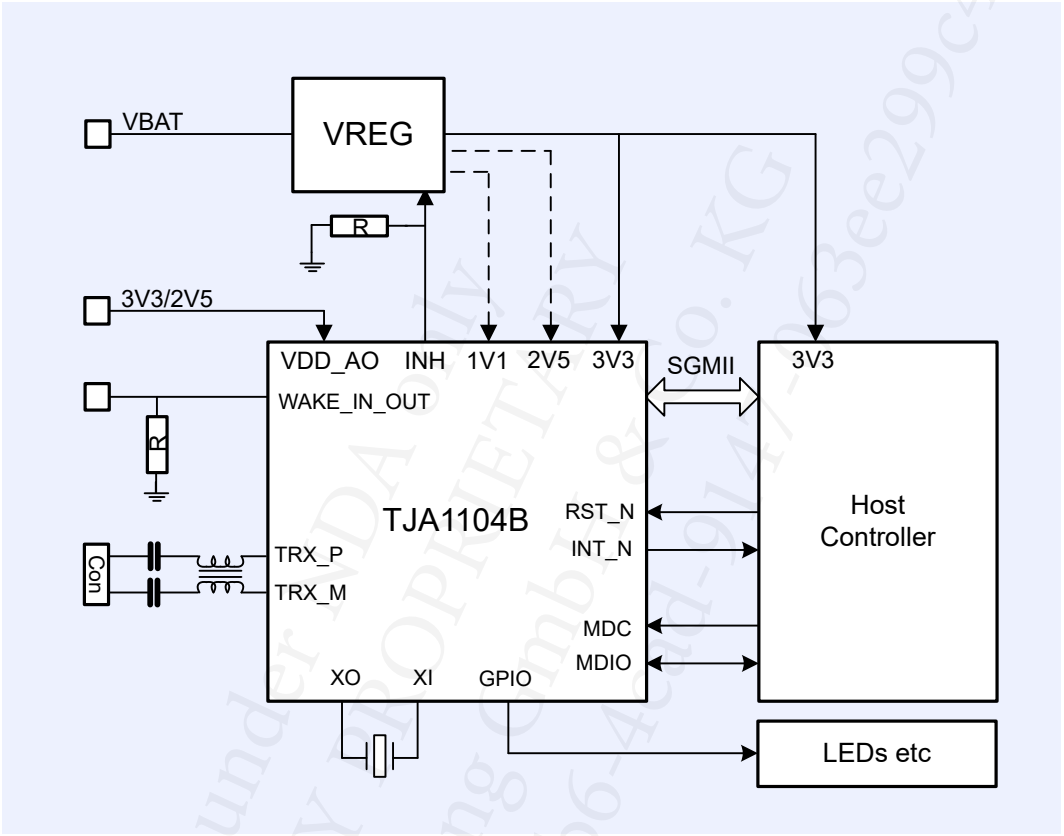


Figure 4. TJA1104B connection diagram

The TJA1104B/D provides the SGMII 4-wire interface which implements clock-data recovery (CDR) so no additional clock pairs are required.

The baud rate of SGMII is fixed at 1.25 Gbaud. For 100 MBit/s data rate, the interface 'elongates' the frame by replicating each octet 10 times.

SGMII must always be AC coupled with a capacitor of 100 nF. The TJA1104B/D is compliant with the SGMII 1.8 specification. The SGMII interface implements (optional) Auto-Negotation. In this mode, PHY and MAC negotiate the supported interface capabilities to find the optimal operating conditions. In SGMII-PHY mode, the device communicates link speed as well as 100BASE-T1 link availability by passing tx_config_Reg from PHY to MAC.

The TJA1104B/D automatically enables the SGMII transmitter (SON, SOP) on device startup in order to reduce time to communication. It is recommended to monitor the SGMII link status through COMPOSITE_STATUS. REPORTED_LINK_STATUS (30.8021h.15) register. With the LINK_STATUS_SELECT bits in the same register, the reported link status can be specified in more detail.

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SGMII (TJA1104B/D only): the SGMII IP encountered a reliablity failure in qualification during HTOL testing. Electrical overstress may cause excessive jitter on the SGMII. Design weakness confirmed by IP supplier. Lifetime or functionality of TJA1104A/C variant is not impacted by this issue.

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3.4 MDI interface

The MDI application circuitry is shown in Figure 5. The common-mode termination depends on OEM requirements and might vary for different OEMs.

The common-mode choke is expected to be compliant with the CMC specification in IEC 62228-5 [4]. The 100 nF coupling-capacitors must have max. 10 % tolerance and ≥50 V voltage rating.

The MDI interface provides ESD robustness of ±6 kV according to IEC61000-4-2 and ±4 kV according to HBM on the IC pins. For further information on the MDI interface, refer to the PCB design guidelines [2]

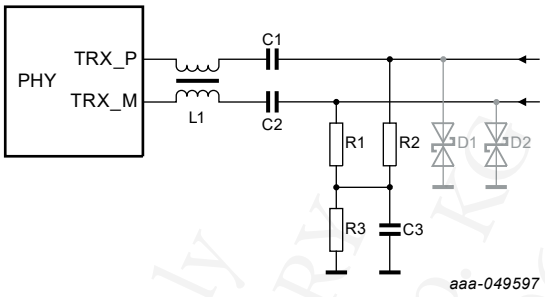


Figure 5. MDI diagram

D1 and D2 shown in Figure 5 are optional.

Table 4. MDI Component specifications

Component	Parameter	Value	Unit
C_1, C_2	Capacitance	100	nF
	Tolerance	≤ 10	%
	Voltage	≥ 50	V
R_1, R_2	Resistance	1	kΩ
	Tolerance	≤ 1	%
	Power rating	≥ 0.4	W
C_3	Capacitance	4.7	nF
	Tolerance	≤ 10	%
	Voltage	≥ 50	V
R_3	Resistance	100	kΩ
	Tolerance	≤ 10	%
	Power rating	≥ 0.1	W
L_1	compliant to IEC 62228-5		
D_1, D_2	ESD protection up to 15kV (compliant to IEC 61000-4-2) [1][2]		

[1] Function of PESD2ETH1G-T was confirmed in internal and external ESD tests.
[2] Function of EZAEG3W11AV was confirmed in internal ESD tests.

3.4.1 Communication channel

The channel used needs to fulfill the channel requirements described in Open Alliance TC2 [5]. Furthermore, to ensure echo is fully canceled, it is recommended to use a channel with a total cable propagation delay of less than 110 ns including the PCB track. Assuming a PCB trace propagation delay of 5 ns for each of the link

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partners this leaves 100 ns propagation delay for the cable. At 6.6 ns/m cable propagation this allows for 15 m cable length. At typical propagation delays of 5 ns/m or lower 15 m can even be exceeded.

3.5 Supply

The TJA1104 offers several supply options including single supply from a 3.3 V or 2.5 V source as well as individual supplies. Furthermore, the supply options for the TJA1104A/C and TJA1104B/D variants are slightly different. The optimal supply configuration depends on the application. In general, however, single-supply operation reduces BOM costs and PCB space while individual supplies allow lower power consumption. The following sections highlight the most common options.

Section 8 provides hints on migrating to the upcoming TJA1120/21 PHYs to help minimize the effort involved. For easiest migration, an individial supply concept as shown in Section 3.5.2 and Section 3.5.4 is recommended.

Table 5 lists specifications for supply filtering components used in Section 3.5.1 and Section 3.5.3 and their respective multiple supply options.

Table 5. Supply component specifications

Component	Parameter	Value	Unit
100 nF	Tolerance[1]	50	%
	Voltage	≥ 6.3	V
1 µF	Tolerance[1]	50	%
	Voltage	≥ 6.3	V
10 µF	Tolerance[1]	50	%
	Voltage	≥ 6.3	V
Ferrite bead	Impedance at 100 MHz	600	Ω
	DC resistance	≤ 0.5	Ω
	Current rating	≥ 300	mA

[1] Over lifetime, including component variation, temperature, aging, etc.

Table 6. Component examples

Component[1]	Manufacturer	Part number
Ferrite beads	Murata	BLM15AG601SN1D
1 µF capacitors	Murata	GRM155R61E105KE11D
100 nF capacitors	TDK	CGA2B3X7R1H104K050BB

[1] Listed set of components was used for EMC measurements.

A detailed overview showing the individual supply currents can be found in the datasheet.

In Table 7, two values are provided for each supply concept: typical and maximum.

- **Typical:** specifies consumption data under typical operating conditions, meaning when the device is active and linked up with random data pattern.
- **Maximum:** as typical but assuming worst-case environment parameters, meaning maximum temperature, voltages on the upper limit and data pattern with worst-case power consumption behavior

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Table 7. Power consumption

Disclaimer: All values are taken from the datasheet. In case of contradictions datasheet values have precedence.

Device/Mode	Parameter	VDDA_SI VDDA_SO	VREGD_IN	VDD_CORE	VREGA_IN	VDDA_RX	VDDA_TRX	VDDA_AO	VDDIO	Total
TJA1104A/C: Single supply MII mode[1]										
Typical	voltage [V]	N/A	3.3	-	3.3	-	-	3.3	3.3	-
	current [mA]	-	17	-	17	-	-	1	14	-
	power [mW]	-	56.1	-	56.1	-	-	3.3	46.2	**161.7**
Maximum	voltage [V]	N/A	3.63	-	3.63	-	-	3.63	3.63	-
	current [mA]	-	37	-	18	-	-	1	14	-
	power [mW]	-	134.3	-	65.34	-	-	3.63	50.82	**254.1**
TJA1104A/C: Individual supply MII mode										
Typical	voltage [V]	N/A	3.3	-	3.3	-	-	3.3	3.3	-
	current [mA]	-	17	-	17	-	-	1	14	-
	power [mW]	-	56.1	-	56.1	-	-	3.3	46.2	**161.7**
Maximum	voltage [V]	N/A	3.63	-	3.63	-	-	3.63	3.63	-
	current [mA]	-	37	-	18	-	-	1	14	-
	power [mW]	-	134.3	-	65.34	-	-	3.63	50.82	**254.1**
TJA1104B/D: Single supply SGMII mode										
Typical	voltage [V]	-	3.3	-	3.3	-	-	3.3	3.3	-
	current [mA]	-	43	-	17	-	-	1	2	-
	power [mW]	-	141.9	-	56.1	-	-	3.3	6.6	**207.9**
Maximum	voltage [V]	N/A	3.63	-	3.63	-	-	3.63	3.63	-
	current [mA]	-	-	-	-	-	-	-	-	-
	power [mW]	-	-	-	-	-	-	-	-	-
TJA1104B/D: Individual supply SGMII mode										
Typical	voltage [V]	N/A	1.1	1.1	2.5	2.5	2.5	3.3	3.3	-
	current [mA]	26	0	16	0	2	15	1	2	-
	power [mW]	28.6	0	17.6	0	5	37.5	3.3	6.6	**98.6**
Maximum	voltage [V]	1.21	1.21	1.21	2.75	2.75	2.75	3.63	3.63	-
	current [mA]	-	-	-	-	-	-	-	-	-
	power [mW]	-	-	-	-	-	-	-	-	-

[1] Current/Power consumption varies slightly for other xMII types, please check datasheet static characteristics for exact numbers

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3.5.1 TJA1104A/C single supply

In the single supply variant, the same source can be used to supply the entire device. Two internal voltage regulators generate the required voltages. Those LDOs can be supplied with 2.5 V or 3.3 V. Depending on the application, the IO voltage can also come from this source. It is possible to supply the device with a 3.3 V or 2.5 V source and run the IO supply on 1.8 V in order to interface low-voltage controllers or reduce emission.

The following schematic shows the minimal single supply bill of material for the TJA1104A. Ferrite beads and capacitors must not be merged into fewer parts. All 100 nF capacitors should be as close to the device as possible followed by the ferrite beads. The pairs of ferrite beads and 1 µF capacitors shall also be placed as close as possible.

Compared with the individual supply circuit presented in Section 3.5.2 the supply filter is optimized for the single supply use-case to minimize the required components. EMC reports are available for both variants.

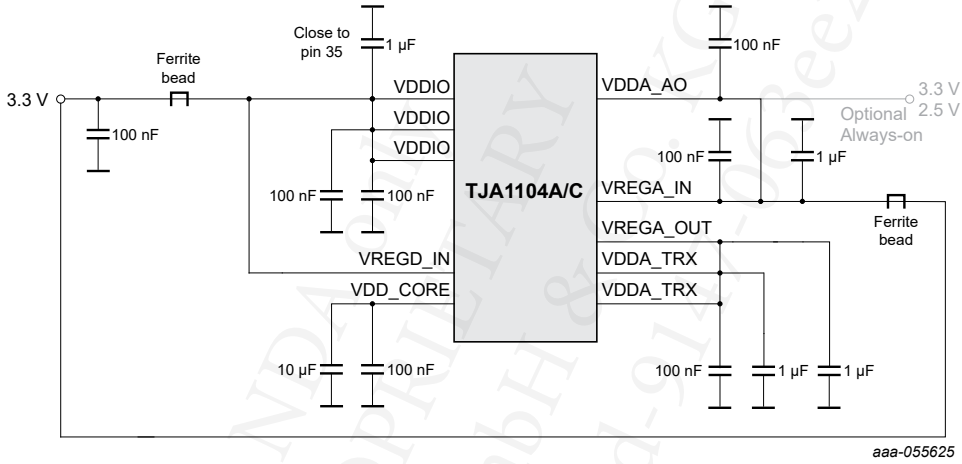


Figure 6. TJA1104A 3.3V single supply circuit

In case the wakeup/sleep functionality is not used, it is recommended to use the supply filtered for VREGA_IN for VDDA_AO as well. When wakeup/sleep is required in the application, VDDA_AO must be present at all times. This includes the Sleep mode. Figure 6 shows how to use the single supply variant in combination with a separate always-on supply. Especially in wake/sleep enabled applications care must be taken to use the correct supply domains for IO pins. See Table 9.

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3.5.2 TJA1104A/C individual supply

When using the individual supply variant, the internal voltage regulators are disabled and bypassed. This is done automatically based on the detected voltages on the VREGD_IN and VREGA_IN pins. The required voltages are shown in the schematics below. The VDDIO supply determines the IO voltage (3.3 V, 2.5 V, or 1.8 V).

All 100 nF capacitors should be as close to the device as possible followed by the ferrite beads. The pairs of ferrite beads and 1 µF capacitors shall also be placed as close as possible. Neither capacitors or ferrite beads must be removed or merged. VDDA_AO is the only supply pin that doesn't require a ferrite bead as filter element.

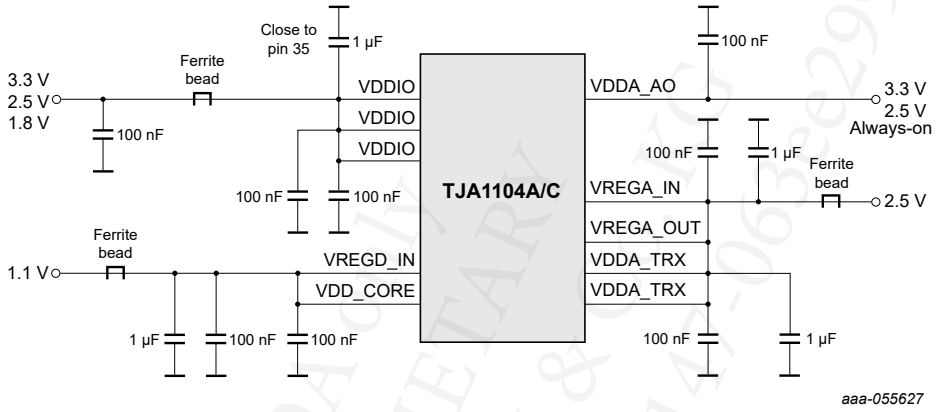


Figure 7. TJA1104A individual supply circuit

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3.5.3 TJA1104B/D single supply

TJA1104B supply variant is identical to TJA1104A except for the supply of the SGMII block which can be seen as VDDA_SI and VDDA_SO in Figure 8. In single supply operation it is possible to source the SGMII block from VDD_CORE which is connected to an internal LDO which itself is supplied by VREGD_IN.

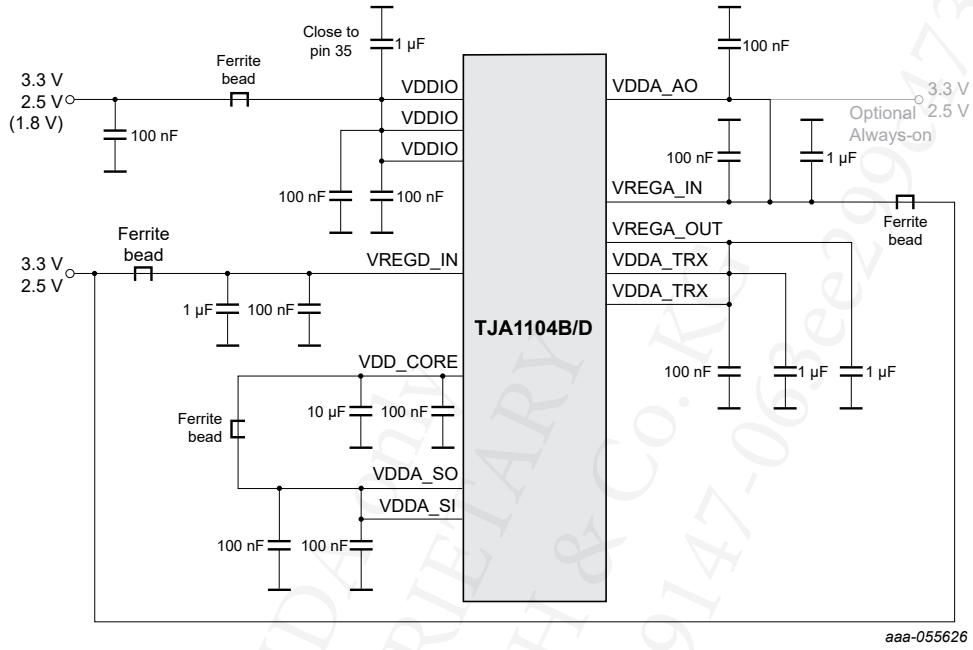


Figure 8. TJA1104B single supply circuit

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3.5.4 TJA1104B/D individual supply

TJA1104B's individual supply concept is identical to TJA1104A's individual supply concept with the additional supply to VDDA_SI / VDDA_SO.

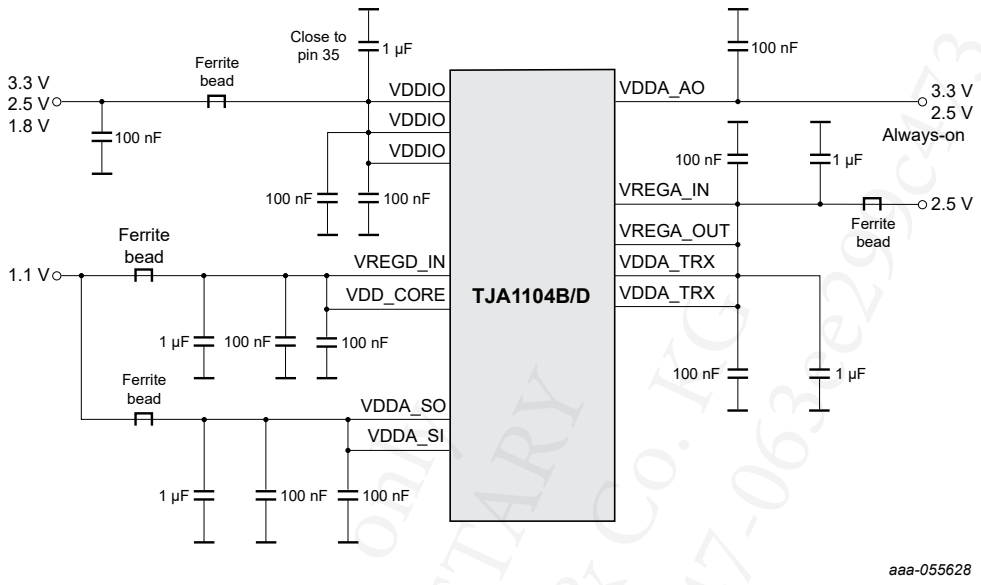


Figure 9. TJA1104B individual supply circuit

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3.5.5 Power-up/down and allowed supply ripple conditions

The TJA1104 has been designed in a way that no dedicated supply sequence is necessary. For instance, it does not matter in which order VDDIO, VDDA_AO, or VREGD_IN, VREGA_IN are started. Being optimized for automotive use cases, the TJA1104 can also handle a wide range of slow and fast ramping supply conditions. See data sheet [1] for details.

Table 8. Permitted ripple per VDD supply

VDD supply	Conditions	Dynamic variation/ripple
3V3 VDDIO		< 50 mV(p-p)
2V2 VDDIO		< 35 mV(p-p)
1V8 VDDIO		< 25 mV(p-p)
VDD33_AO		< 50 mV(p-p)
VREGA_IN	analog supply	< 20 mV(p-p) @ f < 500 kHz
		< 10 mV(p-p) @ f ≥ 500 kHz
VDD25_TRX	2V5 analog supply for 100BASE-T1 (when supplied directly, not using VDDA25_OUT)	< 20 mV(p-p) @ f < 10 MHz
		< 2 mV(p-p) @ f ≥ 10 kHz
VREGD_IN	analog supply	< 50 mV(p-p)
VDD11_CORE	1V1 supply	< 20 mV(p-p)
VDDA11_SGMII	1V1 analog supply	< 20 mV(p-p) @ f < 500 kHz
		< 10 mV(p-p) @ f ≥ 500 kHz

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3.6 Host interface

Figure 10 shows a typical application. Care must be taken to keep the signals in the corresponding supply domain. Alternatively use level shifters. Table 9 gives an overview of the relevant supply domains.

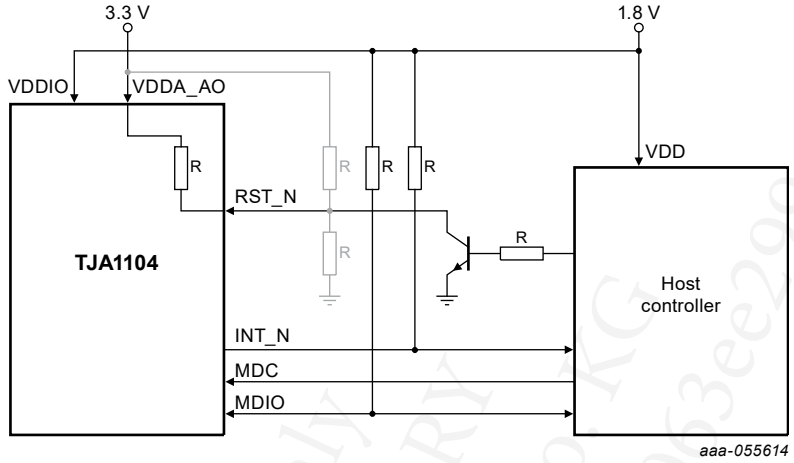


Figure 10. Host interface example VDDIO=1.8V

Table 9. Supply domains

Supply domain	Contained pins
VDDIO	SMI (MDC, MDIO), INT_N, GPIO 0-11[1] xMII (RXD0-3, RXC, RX_CTL, RX_ER, TXD0-3, TXC, TX_CTL, TX_ER)[1]
VDDA_AO	RST_N, WAKE_IN_OUT, INH

[1] Not all pins available depending on configuration

3.6.1 Reset and Interrupt

The reset (RST_N) is an input signal to the PHY and typically driven by a GPIO pin of a microcontroller. The interrupt (INT_N) is an output of the PHY and an input signal to the microcontroller.

If the interrupt is unused, the INT_N pin can be left open.

The reset pin (RST_N) has an internal pull-up of 40-60 kΩ and is pulled up internally when VDDIO is present to guarantee stable pin behavior and no external pull-up or capacitor is needed. This must account for the internal pull-up and other resistors in this line. For example, internal pull-up/pull-down in the microcontroller. See Figure 10 for details.

If the device shall be kept in reset on power-up, it is recommended to add a pulldown resistor of 10 kΩ or less.

RST_N logic levels are according to VDDA_AO.

3.6.2 SMI interface

The SMI (Serial Management Interface) is a serial bus defined in IEEE802.3 Clause 22 and Clause 45. SMI provides access to the various control and status registers for management and service purpose. The TJA1104 supports the Clause 22 protocol version as well as the Clause 45 version. However, Clause 22 access is limited to some basic registers and indirect access to Clause 45 registers.

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SMI communication must only be executed with either Clause 22 or Clause 45 as operating both versions simultaneously[1] is not supported.

In order to access the PHY, the correct PHY address must be known. The PHY address is configured as part of the pin strapping selection and a hardcoded part of the application software.

SMI broadcast through PHY address 0 is disabled by default and can only be enabled via SMI configuration which in turn requires initial knowledge about the PHY address. A typical use case is configuring several PHYs in parallel or changing the operation state of several PHYs simultaneously.

If for some reason the PHY address is not known but only one TJA1104 is present on the bus, a search can be implemented by trying to read the PHY_IDENTIFIER on all possible addresses (1-27).

3.6.2.1 MDIO Pullup

The SMI specification requires a pullup resistor on MDIO. It needs to ensure proper high level when the bus is idle and after every transaction. The recommended value is 1.5 kΩ to 10 kΩ. It must be guaranteed that a high level is reached within one clock cycle. Using the equation for RC circuits (1) and inserting the required timing and threshold conditions (2 and 3) it can be solved for R (4). Table 10 shows some example values under various conditions.

$$U(t) = U_{max}\left(1 - e^{\frac{t}{RC}}\right) \quad (1)$$

$$t = \frac{1}{2}t_{cycle} - t_{setup} \quad (2)$$

$$U(t) = 0.7 \text{ x } U_{max} \quad (3)$$

$$R = \frac{\frac{t_{cycle}}{2} - 10ns}{\ln(0.3)C} \quad (4)$$

Table 10. MDIO pullup resistor configuration

MDIO capacitive load	MDC clock frequency	MDIO pullup value (maximum)	MDIO pullup value (recommended)	Comment
10 pF	f = 2.5 MHz t_{cycle}=400 ns	15.8 kΩ	10 kΩ	Clause 22 speed, short traces, only one PHY connected
100 pF	f = 2.5 MHz t_{cycle}=400 ns	1.5 kΩ	< 1.5 kΩ	Clause 22 speed, long traces and/ or more than one PHY connected
10 pF	f = 10 MHz t_{cycle}=100 ns	3.3 kΩ	1.5 kΩ	High speed, short trace length
50 pF	f = 10 MHz t_{cycle}=100 ns	600 Ω	1 kΩ	High speed, medium trace length **IDLE_PRECHARGE setting required. (see below)**

If the time constant defined by MDIO pullup resistor and trace capacitance cannot be guaranteed to be smaller than one clock cycle, by setting SMI_CONFIG.IDLE_PRECHARGE (MMD30.018Ah.13) the PHY will actively drive MDIO after each read transaction to ensure a correct high level.[2]

1 When switching from Clause 22 to Clause 45 or vice versa perform a hardware reset and stick to the same access scheme until the next hardware reset.

2 It is recommended to enable this setting only when needed.W

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3.6.2.2 IEEE 802.3 Clause 45

For a full Clause 45 register access, two transactions are required. First an address frame is sent to specify the MMD (MDIO Managed Device Number) and register address. Then, a data frame is sent to perform the data read or write.

In addition to this, there exists a read operation that increments the register address which allows subsequent reads without sending address frames. The address pointer is kept individual for each MMD.

Table 11. Management frame fields

Frame	PRE	ST	OP	PHYADDR	DEVAD	TA	Adress/Data	Idle
Address	1...1	00	00	PPPPP	EEEEE	10	AAAAAAAAAAAAAAAA	Z
Write	1...1	00	01	PPPPP	EEEEE	10	DDDDDDDDDDDDDDDD	Z
Read	1...1	00	11	PPPPP	EEEEE	Z0	DDDDDDDDDDDDDDDD	Z
Post-read increment address	1...1	00	10	PPPPP	EEEEE	Z0	DDDDDDDDDDDDDDDD	Z

- ST = Start-of-Frame = ‘00’ (fixed) to define Clause 45 data frames
- OP = Operation code (’1x’ = Read, ’0x’ = Write)
- PRTAD = 5-bit Port address (MSB first) = PHY Address
- DEVAD = 5-bit (MMD)
- ADDR/DATA = 16-bit Address or Register value (bit 15 first)

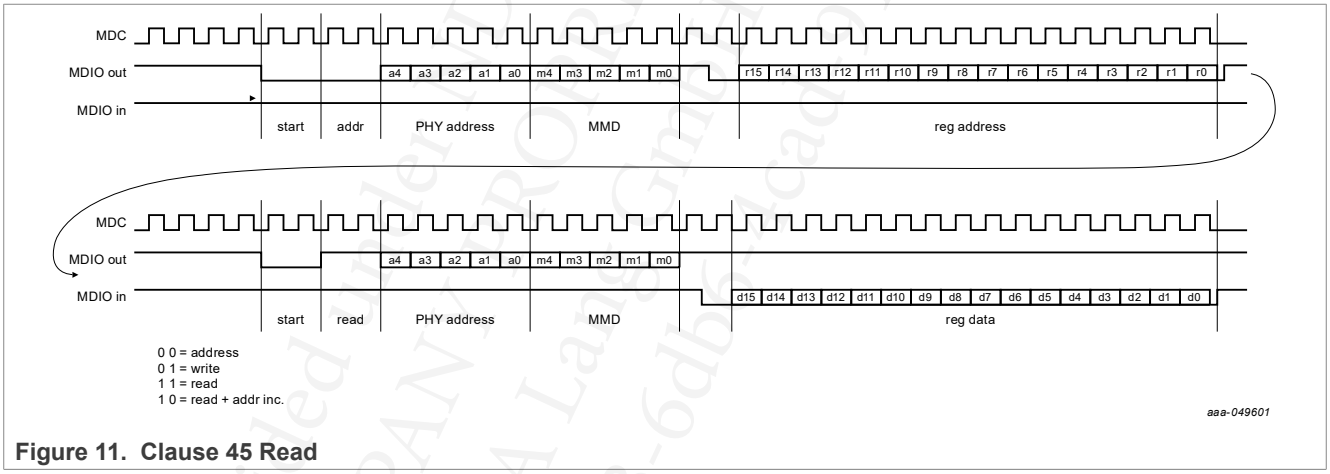


Figure 11. Clause 45 Read

In case multiple consecutive registers shall be read instead of repeating the address phase for every read, a read+address increment operation can be used. This doubles the throughput. An example is shown below.

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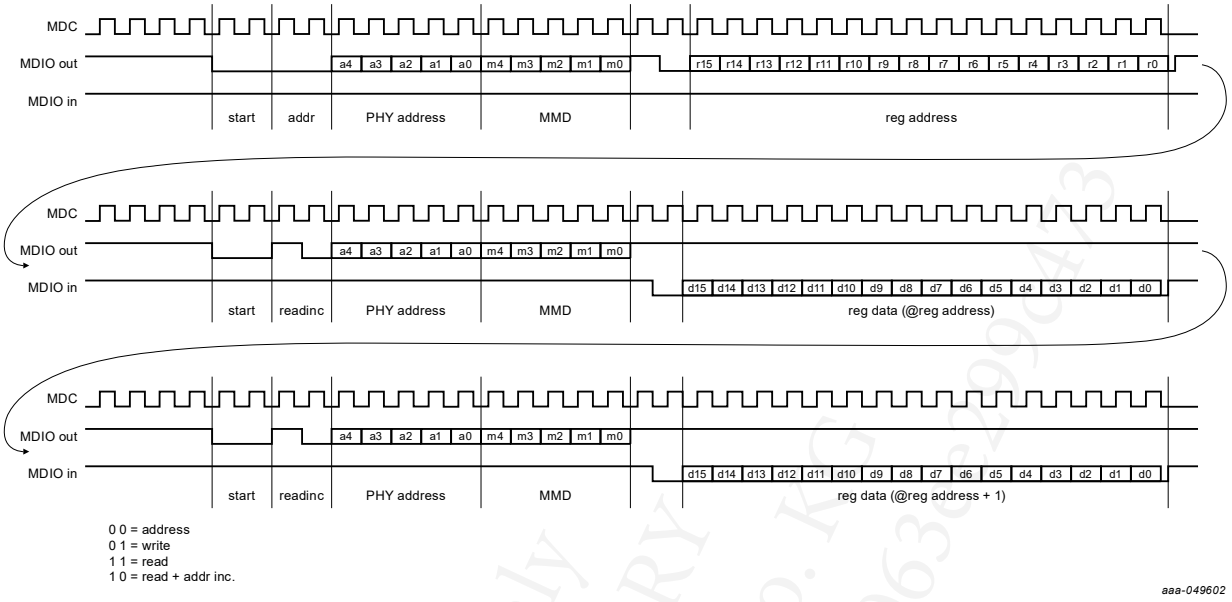


Figure 12. Clause 45 Read with address increment

3.6.2.3 IEEE 802.3 Clause 22

For Clause 22 register access, a single SMI frame specifies both address and data to be read or written. Figure 13 and Figure 14 show timing diagrams for read and write.

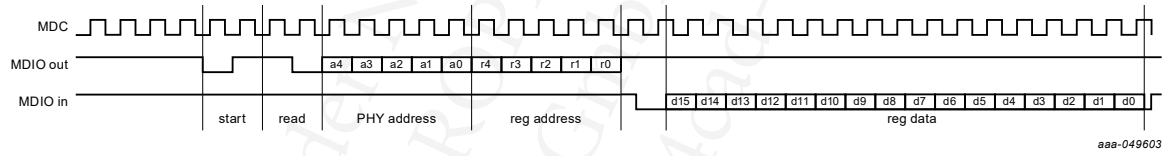


Figure 13. Clause 22 Read

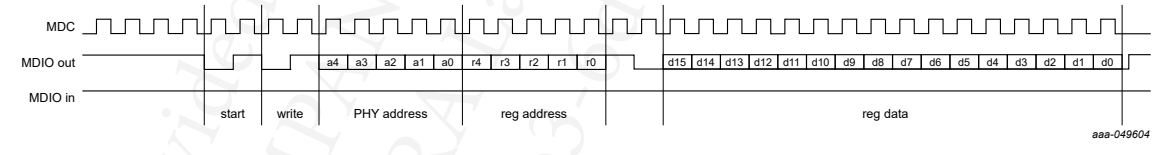


Figure 14. Clause 22 Write

Via Clause 22, the TJA1104 only allows direct access to 4 device registers as well as 3 additional registers that allow indirect access to Clause 45 registers.

The registers available under Clause 22 are:

- PHY_ID_1
- PHY_ID_2
- FAVORITE_IRQS
- ALWAYS_ACCESSIBLE

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- CL45_ACCESS_CONTROL[3]
- CL45_ADDRESS_DATA[4]
- CL45_ADDRESS[5]

FAVORITE_IRQS is a shortcut to various bits in the IRQ tree. All data presented here can also be found in the respective *_IRQ_MSTATUS register. ALWAYS_ACCESSIBLE is the only register accessible without the presence of VDD_CORE. It signals whether other registers are available for read and write and can be used to wake up the device from sleep. In addition, it can be used to read the status of various monitors.

The registers that allow access to Clause 45 registers are described in the Section 3.6.2.3.1.

Other registers reside in the Clause 45 space. In case the MAC does not support Clause 45, refer to Section 3.6.2.3.1.

3.6.2.3.1 Clause 22 to Clause 45 indirect addressing

There are registers in the Clause 22 space (CL45_ADDRESS, CL45_ADDRESS_DATA and CL45_ACCESS_CONTROL) which allow an indirect access scheme to read or write registers in the Clause 45 address space. For a complete Clause 45 transaction four steps must be performed:

1. Select MMD and set OP=address in CL45_ACCESS_CONTROL
2. Write Clause 45 register address to CL45_ADDRESS_DATA
3. Select MMD and set OP=data in CL45_ACCESS_CONTROL
4. Write or read register content to/from CL45_ADDRESS_DATA

Writes or reads to CL45_ADDRESS_DATA trigger the operation specified in CL45_ACCESS_CONTROL so the order of access must be followed.

The following listing shows an example reading PHY_CONFIG:

```
//write MMD=30+OP=address to CL45_ACCESS_CONTROL
write 13 0b00xxxxxxxx11110

//write PHY_CONFIG addr to CL45_ADDRESS_DATA
write 14 8108h

//write MMD=30+OP=data to CL45_ACCESS_CONTROL
write 13 0b01xxxxxxxx11110

//read data from CL45_ADDRESS_DATA corresponding to 30.8108h
read 14
```

In order to speed up the sequence shown a third register can be used to change the address without the need of changing the operation in CL45_ACCESS_CONTROL:

1. Select MMD and set OP=data in CL45_ACCESS_CONTROL
2. Write Clause 45 register address to CL45_ADDRESS
3. Write or read register content to/from CL45_ADDRESS_DATA

If subsequent accesses fall into the same MMD, the write to CL45_ACCESS_CONTROL can be omitted which reduces the number of clause 22 transactions from 4 to 2.

3 Used for clause 22 to clause 45 indirect addressing
4 Used for clause 22 to clause 45 indirect addressing
5 Used for clause 22 to clause 45 indirect addressing

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The following listing shows an example reading PHY_CONFIG followed by a write to another register in the same MMD

```
//write MMD=30+OP=data to CL45_ACCESS_CONTROL
write 13 0b01xxxxxxxxx11110

//write PHY_CONFIG addr to CL45_ADDRESS
write 30 8108h

//read data from CL45_ADDRESS_DATA corresponding to 30.8108h
read 14

//write a different addr to CL45_ADDRESS
write 14 8100h

//write data to the register
write 14 0bxxxxxxxxxx000100
```

3.6.2.4 Mixing Clause 22 and Clause 45 access

When the device is woken up under Clause 45 via the ALWAY_ACCESSIBLE register (address Shared.801Fh) and the next SMI access is performed under Clause 22, the wrong data will be returned.

Workaround: If clause 45 is used to wake up the device, make at least one more clause 45 access to a different register before switching to Clause 22. Switching from clause 22 to clause 45 is not affected.

3.6.2.5 Additional SMI handling

SMI_ERROR (ALWAYS_ACCESSIBLE)
Reading the ALWAYS_ACCESSIBLE register (address Shared.801Fh) using Clause 22 will wrongly set bit SMI_ERROR in the EPHY_CONFIG_REGXS_IRQ_SOURCES register (address Shared.80C0h.1).
Workaround: Ignore/reset bit SMI_ERROR after reading the ALWAYS_ACCESSIBLE register.

SMI_ERROR (reserved MMD)
A write transaction to a reserved MMD will (wrongly) not set bit SMI_ERROR in EPHY_CONFIG_REGXS_IRQ_SOURCES (address Shared.80C0h.1). Read transactions are not affected.
Workaround: None

SMI_ERROR (CL22 wrong opcode)
When a wrong opcode is observed in a CL22 SMI transaction, the next transaction will wrongly cause bit SMI_ERROR in EPHY_CONFIG_REGXS_IRQ_SOURCES (address Shared.80C0h.1) to be set.
Workaround: Ignore/reset bit SMI_ERROR.

3.7 WAKE_IN_OUT Pin

The bidirectional WAKE_IN_OUT pin can be used as local wake-up input or wake-up output. The wakeup-up output is intended to forward a wake-up to another PHY or switch. The input logic is positive. Thus, a rising edge triggers a wake-up event. The output logic is open-drain. When inactive the pin is floating. When generating a wake-up pulse, the pin is pulled to VDDA_AO.

An external pull-down resistor is required and the pin must not be floating. A value between 10 kΩ and 100 kΩ is recommended.

The value of the resistor depends on the total capacitance and the desired fall time. The leakage current of the input is less than 1 µA and can be ignored. The input capacitance of the pin is less than 5 pF. If a level of

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0.35×VDDIO is required as low threshold level, a simple RC low-pass calculation can be used: Assuming a total capacitance of 50 pF and a desired fall time of 5 µs the resistor value must be 100 kΩ

The actual behavior of the pin can be configured via WISE_CONFIG.WU_IO (30.0182h 12:13). The following table shows the supported options:

Table 12. Wake-up IO configuration - WU_IO_ENABLE

Bits 12:13	Description
00	Wake-up IO is disabled
01	Wake-up IO is input only, output disabled
10	Wake-up IO is output only, input disabled
11	Wake-up IO is input and output

The length of a generated wake-up pulse can be configured via WISE_PARAMETERS.WU_IP_PULSE_LENGTH.(30.0184h.3:0)

3.8 INH Pin

The intention of INH is to control voltage regulators within the ECU. In Figure 3 the voltage regulator for the host controller and the PHY is enabled via the INH output of the TJA1104.

The pin INH provides a VDDA_AO related high-side switch which is floating during sleep mode. This design allows combining INH/EN pins of multiple devices to achieve a wired-or topology. Thus if another device is not actively driving INH high, a LOW signal on the inhibit input disables the regulator.

An external pull-down or a pull-down embedded in the voltage regulator is needed to ensure a low level. The recommended value is 5 kohms to 20 kohms. This is shown in Figure 15.

If the TJA1104 is not in sleep, the pin INH is switched to the VDDA_AO voltage. This indicates that the voltage regulator shall be enabled.

If the wakeup/sleep feature is unused and the INH pin is not used, it can be left open.

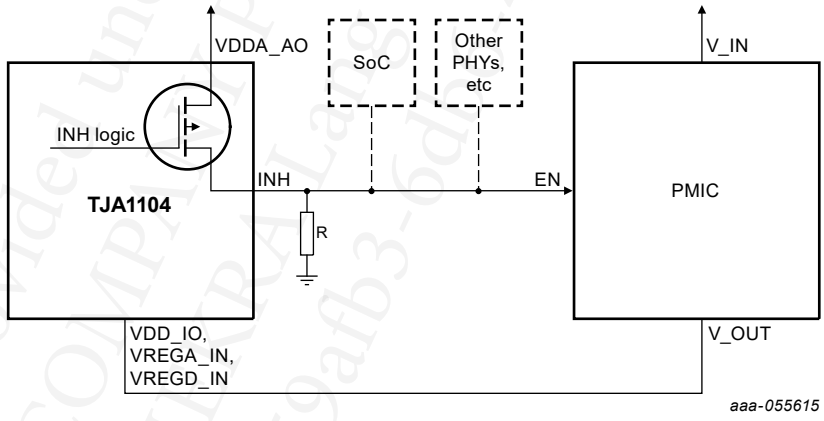


Figure 15. INH topology example

3.8.1 Voltage level adaption

In case the PMIC has a higher enable threshold than $V_{DDA(AO)}$ a voltage level translation must be implemented. Figure 16 shows an example circuit with component values listed below.

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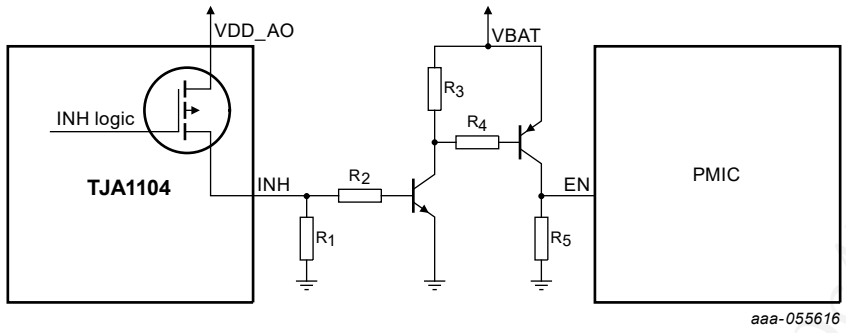


Figure 16. INH voltage level adaption

Table 13. INH translation circuit component specifications

Component	Parameter	Value	Unit
$V_{DDA(AO)}$	Voltage	3.3	V
V_{BAT}	Voltage	12	V
R_1	Resistance	10	kΩ
R_2	Resistance	68	kΩ
R_3	Resistance	100	kΩ
R_4	Resistance	68	kΩ
R_5	Resistance	4.7	kΩ
Q_1	Type	BXC70C	
Q_2	Type	BCX71C	

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3.9 Clocking

There are different possibilites to clock the TJA1104: A 25MHz crystal, the 50 MHz REF_CLK input of the RMII interface or a single-ended clock input.

In case a crystal is used, it must fulfill the requirements as given in Table 14.

Table 14. XTAL requirements

Parameter	Requirement
Frequency	25 MHz
Tolerance	50 / 100 ppm [1][2]
ESR	max 100 Ω
Total load capacitance	8 pF, see description below

[1] For details, see Section 3.9.3
[2] Over lifetime and operating conditions

Table 15. Oscillator characteristics

Parameter	Min	Typ.	Max	Unit
DC transconductance (gm)	2	15	32	mA/V
Drive level			100	µW

3.9.1 Schematics

The schematic in Figure 17 gives an example for a typical application. R_{S1} and R_{S2} are optional and may be added to limit the drive strength and improve EMI robustness.

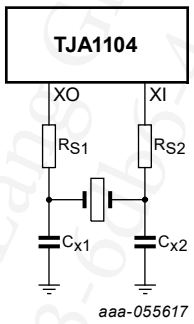


Figure 17. XTAL schematics

Table 16. clocking circuit component example

Component	Parameter	Value	Unit
R_{S1} / R_{S2}	Resistance	15[1]	Ω
C_{x1} / C_{x2}	Capacitance	10[2][3].	pF

[1] Recommended for increased EMC robustness.
[2] Recommended when using NX3225SA-25.000M-STD-CRS-2.
[3] Recommended when using XRCGB25M000F3A00R02

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3.9.2 Load capacitance and drive

Every oscillation mode of the crystal is represented by an inductance, capacitance, and resistance in series. The fundamental frequency is represented by L_1, C_1 and R_1. All series branches are shunted by the parallel package capacitance C_p.

The equivalence model of the crystal in application is shown in Figure 18.

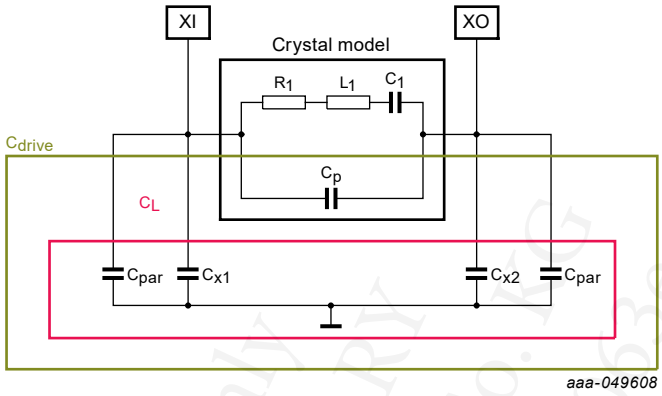


Figure 18. Crystal and application equivalence model

C_L is the typical load capacitance of the crystal and is specified by the crystal manufacturer. When using a crystal that is manufactured for a different load capacitance, the circuit oscillates at a slightly different frequency compared to the specified one.

The total load capacitance consists of PCB parasitics capacitance, the shunt capacitance of the crystal and the discrete load capacitors C_{x1} and C_{x2}.

$$C_L = C_{par} 2 + \frac{C_{x1} \cdot C_{x2}}{C_{x1} + C_{x2}}$$

The value of C_x can be calculated as follows:

$$C_{x1} = C_{x2} = (2 \cdot C_L) - C_{par}$$

The parasitics depend on the PCB layout and the TJA1104's pin capacitance. As a rule of thumb, use C_x=2×C_L - 4 pF.

Note: *Always validate the ppm accuracy of the application. Ethernet demands a stable frequency with a ±100 ppm limits. The frequency is influenced by the board layout and selected components.*

The power dissipated in the crystal is called the drive level. In general crystal characteristics are specified at drive levels of 1 µW to 0.5 mW. Higher drive levels (between 1 mW and 5 mW) can accelerate ageing or may change the crystal's physical parameters permanently. The level of drive can be calculated using equation:

$$P_{drive} = R_1 \ (2\pi f v_{eff} C_{drive})^2$$

In which R_1 is the series resistance of the crystal, f is the oscillation frequency, v_{eff} effective value of the voltage across the crystal. which is $1/\sqrt{2}$ of the amplitude. C_{drive} is the total differential capacitance C_{drive}=C_L+C_P. The amplitude under typical conditions is 1.1 V.

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3.9.3 Frequency stability

An accurate clock is important in Ethernet communication. Many Ethernet subsystems make assumptions on the clock frequency and its stability in order to operate properly. The IEEE 802.3 standard mandates a clock accuracy of 100 ppm.

In the xMII interface of the TJA1104 elastic buffers are used. They compensate between clock differences of the 100BASE-T1 link and the xMII interface. These buffers are able to compensate minimal data rate differences assuming a worst-case difference of 200 ppm. The exact operation depends on the selected xMII mode and is explained in the TJA1104 Datasheet [1].

The RMII and RGMII interface specifications are an exception to this. By standard, these interfaces operate on clocks of ±50 ppm clock accuracy. In order to reduce BOM cost, the TJA1104 is able to operate with 100 ppm clocks in all interface modes. This device can operate with a 100 ppm crystal and also accept 100 ppm xMII interface clocks in all modes of operation. As a consequence, the PHY generates a clock of 100 ppm to the MAC.

Note: *For RMII and RGMII it must be confirmed that the MAC also supports 100 ppm interface clocks. If not, it is recommended to use a ±50 ppm crystal in RMII or RGMII mode.*

3.9.4 External clock

In case the MAC or another device is able to output a 25 MHz clock it can be used to clock the TJA1104. This is done by connecting the clock to the XI pin of the oscillator. The input of the oscillator is self-biasing and the external clock must be AC coupled. For this, a 100 pF capacitor is recommended. The XI input is damaged if voltages larger than 1.21 V (p-p) are applied. The peak-to-peak voltage can be in the window of 0.6 - 1.21 V (p-p) but it recommended to be as high as possible. For example 1.1 V (p-p) which allows for over-shoot margin. A large swing generally reduces jitter.

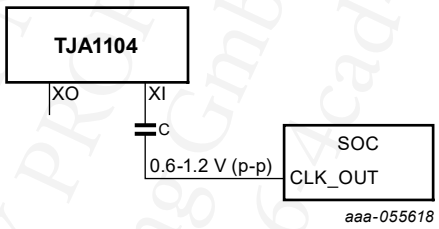


Figure 19. External clock diagram

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3.10 Pin Strapping

The TJA1104 allows for basic hardware configuration via pin-strapping during power-on. The following pins are used in several function blocks:

Table 17. Pinstrap pin to function mapping

Pin #	Function
17, 19, 20	PHY address
21, 23	xMII mode
24, 25	Misc (MASTER/SLAVE/AUTONOMOUS)

Each pin can have 3 states as listed in the table below. The pull-up/down resistors should have a value of 10 kΩ. When using the config pins as GPIOs, for example as an LED output, the LED polarity must be matched to the pin strapping resistor. Alternatively a FET must be used to detach the load from the pin strapping mechanism. Additional (parasitic) impedance that conflicts with the intended pin strap state must be avoided.

Table 18. 3-Level Strap Resistor Values

Setting	Pull-Up Resistor	Pull-Down Resistor
O	Open	Open
H	10 kΩ	Open
L	Open	10 kΩ

The state of the pin strap is sampled at device startup or device reset. The result of the pin strapping is latched in registers accessible via SMI. All settings can be read and modified (except the PHY address) through MDI.

Figure 20 shows several different pinstrap options. The shown setup will result in CONFIG0-6 = HOH OO OL[6] which configures the PHY to ADDR=7 Mode=RGMII-ID Slave/Autonomous/Polarity correction enabled.

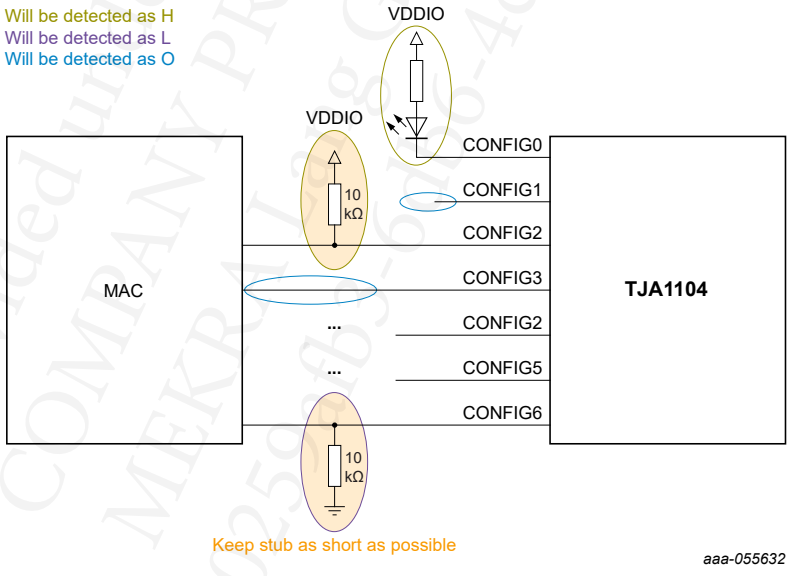


Figure 20. Pin strapping Hardware

6 Assuming config pins not shown are not connected or connected in the same way as CONFIG3.

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3.10.1 Detection logic

The actual detection takes place in two phases:

1. Enable internal pulldown (disable pullup) and sample pin
2. Enable internal pullup (disable pulldown) and sample pin

The four possible results are decoded to the effective pin strap configuration as follows:

Table 19. Pinstrap decoding

Sample result pulldown phase	Sample result pullup phase	Decoded result	Pinstrap result
0	0	External pulldown	L
1	1	External pullup	H
0	1	Floating	O
1	0	Invalid	O[1]

[1] The startup is not interrupted so a valid result has to be assumed

Figure 21 shows a simplified version of the IO-cell. The colored boxes contain the resistors relevant for the three different configuration options. The value of the internal pullup/pulldown resistors is between 40 kΩ - 62 kΩ which leads to the following limits for the external resistors:

Table 20. Pinstrap constraints

Desired value	Detection threshold requirement	Constraint
H	V_{IO} > VDDIO×0.7	R_{pullup} < 17 kΩ
L	V_{IO} < VDDIO×0.3	$R_{pulldown}$ < 17 kΩ
O	V_{IO} > VDDIO×0.7 with internal PU V_{IO} < VDDIO×0.3 with internal PD	RP_{pullup} > 145 kΩ $RP_{pulldown}$ > 145 kΩ

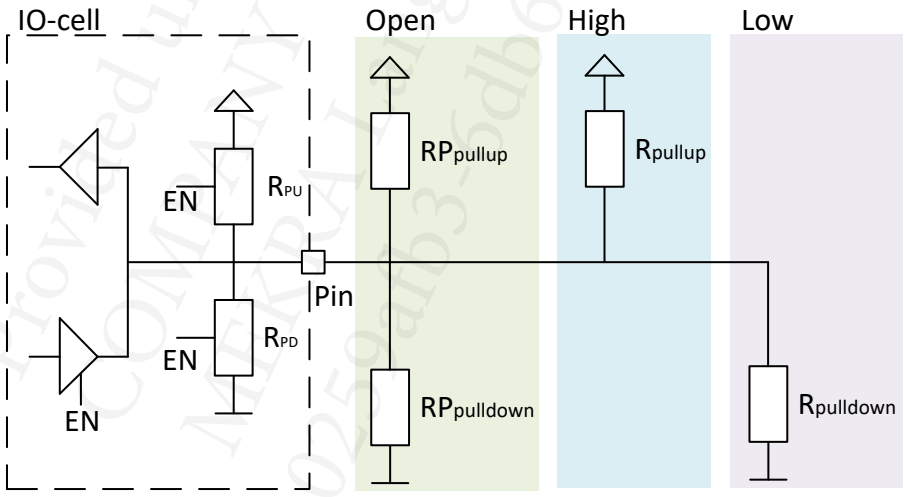


Figure 21. Simplified IO-Cell

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3.10.2 xMII Mode Configuration

Two configuration pins determine the default xMII mode. The possible options and corresponding pin strap settings are listed in Table 21. TJA1104 allows reconfiguration of xMII interface through register settings which is described under Section 4.4.

The TJA1104A supports CMOS-type MII/RMII/RGMII modes and the TJA1104B supports SerDes SGMII.

Table 21. Pinstrap MODE group

xMII Mode (TJA1104A)	xMII Mode (TJA1104B)	Pin CONFIG3	Pin CONFIG4
RGMII-ID (RXC)	SGMII-PHY	O	O
RGMII	SGMII-MAC	O	H
RGMII-ID (both TXC/RXC)		O	L
RMII		H	O
RMII-clkmode		H	H
RMII-rev		H	L
MII		L	O
MII-rev		L	H
JTAG	JTAG	L	L

3.10.3 Miscellaneous Configuration

Pins CONFIG5 and CONFIG6 control the following miscellaneous settings.

- MASTER/SLAVE
- AUTONOMOUS mode
- Polarity correction

The AUTONOMOUS operation mode and the MASTER/SLAVE settings are explained in Section 2.2. It is adviced to enable the polarity correction.

In case the TJA1104 is pin strapped for Managed mode operation, the PHY pauses at start-up, until the DEVICE_CONTROL.START_OPERATION bit (30.0040h.0) is set.

Table 22. Pinstrap MODE group

MASTER/SLAVE	AUTONOMOUS Mode	Polarity Correction	Pin CONFIG5	Pin CONFIG6
MASTER	enabled	enabled	O	O
MASTER	disabled	enabled	O	H
SLAVE	enabled	enabled	O	L
SLAVE	disabled	enabled	H	O
MASTER	enabled	disabled	H	H
MASTER	disabled	disabled	H	L
SLAVE	enabled	disabled	L	O
SLAVE	disabled	disabled	L	H
reserved	*reserved*	*reserved*	L	L

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3.10.4 PHY Address Configuration

The PHY address is used for SMI addressing, see also Section 3.6.2. PHY addresses 1 to 27 can be selected via 3 pin strap pins. For details refer to the Datasheet [1]. Figure 22 shows an example for PHY address = 2. As the PHY address is used in the SMI communication itself, it is not possible to change the PHY address via SMI access.

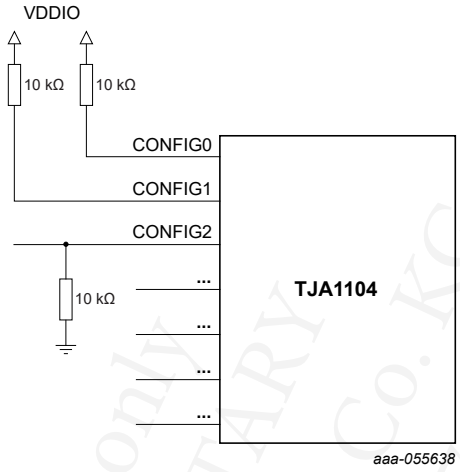


Figure 22. PHY Address config example

3.11 IO pins

TJA1104 has 2 dedicated general purpose I/Os pins and, depending on the xMII configuration, up to 6 pins that can optionally be configured as GPIOs. Please refer to section 6.13.2 in the datasheet for details on which GPIO is available in which xMII mode and how to configure it.

The default configuration for all GPIOs is input with the exception of GPIO0 which is configured as link status LED. In order to manually configure a GPIO as input, FUNC_SELECT & SIGNAL_SELECT of the corresponding GPIOx_FUNC_CONFIG register must be set to MISC (3b111) and SNIF(5b00000) respectively.

3.11.1 IO driver configuration

The drive parameters of the IO drivers can be configured for optimal supply- and application condition. The TJA1104 automatically selects a predefined default setting based on the detected VDDIO level which avoids the need to change these values for typical applications. [7] The default preset caters for up to 25 pF trace and far end load. In case the PHY is close to the MAC an ALTERNATE_CONFIG preset can be set. The ALTERNATE_CONFIG is optimized for up to 15 pF and improves the EMC behavior. This setting is only available for IO cells which are part of the xMII interface. It can be found in the register XMII_DATA_IO_CONFIG.

The driver settings for various IO cell groups can be found in the following registers.

Note: *It is strongly recommended to either use the default oder ALTERNATE_CONFIG. Deviating from these predefined values can impact signal integrity and EMC characteristics and require careful investigations. Timing values listed in the datasheet are valid only for the predefined settings.*

7 The predefined value cannot be read back via SMI, see description in Linktext-Table_

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Table 23. IO driver configuration

Register	Pins / Function
SMI_DATA_IO_CONFIG	MDIO pin
XMII_CLK_IO_CONFIG	MII: TX_CLK, RX_CLK revRMII: REF_CLK RGMII: RXC
XMII_DATA_IO_CONFIG	MII/RMII/RGMII RXD and TXD pins
GPIO_IO_CLK_CONFIG	GPIO9 when configured for REF_CLK_OUT, TX_CLK_OUT modes
GPIO_IO_CONFIG	All other GPIO function

The register layout is the same for all function blocks and contains the following settings:

Table 24. Register layout

Bit	Symbol	Description
15	MANUAL_CONFIG	If set, custom values are used to set configuration. Otherwise default values are used
14	ALTERNATE_CONFIG	Only for XMII_DATA_IO_CONFIG If set, reduced emission selection for XMII data and clock pads. (up to 15 pF load)
10:8	DRIVE_STRENGTH[1]	Select drive strength of IOs 0 = weakest 7 = strongest
2:0	SLEW_RATE[1]	Select slew rate of IOs 0 = weakest 7 = strongest
others	*reserved*	Read/Write as 0

[1] The power on value is 0. In case MANUAL_CONFIG = 0 the value in this field does not reflect the setting actually used.

Table 25 shows the predefined default IO-cell parameters for each IO-cell group. Each column represents one group. The groups correspond to the registers listed in Table 23. There is no dedicated column for GPIO_IO_CLK_CONFIG as the predefined default values are identical to GPIO_IO_CONFIG. Please note that the relation between pin and IO-cell group is dynamically set based on the xMII mode and pin configurations.

Note: *This table only applies for TJA1104A/C. On TJA1104B/D all GPIO pins are in the GPIO group and subsequently controlled by the register GPIO_IO_CONFIG.*

Table 25. Predefined default values TJA1104A/C

	SMI	GPIO	XMII_DATA	XMII_CLK
Default 1.8V	DRV_STRENGTH=5 SLEW_RATE=7	DRV_STRENGTH=5 SLEW_RATE=4	DRV_STRENGTH=5 SLEW_RATE=7	DRV_STRENGTH=5 SLEW_RATE=7
Default 2.5V	DRV_STRENGTH=4 SLEW_RATE=5	DRV_STRENGTH=4 SLEW_RATE=4	DRV_STRENGTH=4 SLEW_RATE=5	DRV_STRENGTH=4 SLEW_RATE=5
Default 3.3V	DRV_STRENGTH=4 SLEW_RATE=5	DRV_STRENGTH=4 SLEW_RATE=4	DRV_STRENGTH=4 SLEW_RATE=5	DRV_STRENGTH=4 SLEW_RATE=5
Default 1.8V ALTERNATE_CONFIG	-	-	DRV_STRENGTH=5 SLEW_RATE=7	DRV_STRENGTH=5 SLEW_RATE=7

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Table 25. Predefined default values TJA1104A/C*...continued*

	SMI	GPIO	XMII_DATA	XMII_CLK
Default 2.5V ALTERNATE_CONFIG	-	-	DRV_STRENGTH=4 SLEW_RATE=5	DRV_STRENGTH=4 SLEW_RATE=5
Default 3.3V ALTERNATE_CONFIG	-	-	DRV_STRENGTH=4 SLEW_RATE=5	DRV_STRENGTH=4 SLEW_RATE=5
Pins MII	36 MDIO	17 GPIO0 18 GPIO1 19 GPIO2[1] 28 GPIO9[2]	19 RX_ER[3] 20 RXD3 21 RXD2 23 RXD1 24 RXD0 25 RX_CTL	26 RXC 29 TXC
Pins RMII	36 MDIO	17 GPIO0 18 GPIO1 19 GPIO2[1] 28 GPIO9[2] 33 GPIO10 34 GPIO11 20 GPIO3 21 GPIO4 26 GPIO8	19 RX_ER[3] 23 RXD1 24 RXD0 25 RX_CTL	29 TXC
Pins RMII-alt[4]	36 MDIO	17 GPIO0 18 GPIO1 19 GPIO2[1] 28 GPIO9[2] 33 GPIO10 34 GPIO11 21 GPIO3 23 GPIO4 26 GPIO8	19 RX_ER[3] 24 RXD1 25 RXD0 20 RX_CTL	29 TXC
Pins RGMII[5]	36 MDIO	17 GPIO0 18 GPIO1 19 GPIO2 28 GPIO9	20 RXD3 21 RXD2 23 RXD1 24 RXD0 25 RX_CTL	26 RXC

[1] Optional, if not used as RX_ER
[2] Optional, if not used as TX_ER
[3] Optional, if not used GPIO2 is available
[4] Alternate xMII arragement shown in section 6.4.5 in the datasheet
[5] Alternate xMII arragement does not change IO-cell configuration grouping

Table 26. Predefined default values TJA1104B/D

	SMI	GPIO	XMII_DATA	XMII_CLK
Default 1.8V	DRV_STRENGTH=5 SLEW_RATE=7	DRV_STRENGTH=5 SLEW_RATE=4	DRV_STRENGTH=5 SLEW_RATE=7	DRV_STRENGTH=5 SLEW_RATE=7
Default 2.5V	DRV_STRENGTH=4 SLEW_RATE=5	DRV_STRENGTH=4 SLEW_RATE=4	DRV_STRENGTH=4 SLEW_RATE=5	DRV_STRENGTH=4 SLEW_RATE=5

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Table 26. Predefined default values TJA1104B/D*...continued*

	SMI	GPIO	XMII_DATA	XMII_CLK
Default 3.3V	DRV_STRENGTH=4 SLEW_RATE=5	DRV_STRENGTH=4 SLEW_RATE=4	DRV_STRENGTH=4 SLEW_RATE=5	DRV_STRENGTH=4 SLEW_RATE=5
Default 1.8V ALTERNATE_CONFIG	-	-	DRV_STRENGTH=5 SLEW_RATE=7	DRV_STRENGTH=5 SLEW_RATE=7
Default 2.5V ALTERNATE_CONFIG	-	-	DRV_STRENGTH=4 SLEW_RATE=5	DRV_STRENGTH=4 SLEW_RATE=5
Default 3.3V ALTERNATE_CONFIG	-	-	DRV_STRENGTH=4 SLEW_RATE=5	DRV_STRENGTH=4 SLEW_RATE=5
SGMII pins	36 MDIO	17 GPIO0 18 GPIO1 19 GPIO2 28 GPIO9	30 SON 31 SOP 32 SIP 33 SIN	-

In addition to the listed default values, Table 27 shows the values available for xMII data pins for different VDDIO levels.

Blue fields are available modes for 3.3 V only. Fields marked green are available modes for 2.5 V and 3.3 V. Fields in orange are the available modes for 1.8 V only.

It is not recommended to go to a more aggressive (higher DRIVE_STRENGTH value) setting. There is little improvement in the timing but there is a reduction in the driver output impedance which leads to undesirable emission.

Table 27. Setting for 15 pF/25 pF mode (ALTERNATE_CONFIG = 1/0): TJA1104

25pF	DRIVE_STRENGTH								
		0	1	2	3	4	5	6	7
SLEW_RATE	0-4								
	5					Default 2.5 / Default 3.3			
	6								
	7						Default 1.8		

3.12 JTAG

JTAG can be used to test all IOs available on the TJA1104.

In order to utilize JTAG the interface must be activated via pinstrapping. One solution is to let the host controller dynamically configure pinstrapping before switching the corresponding pins to xMII function.

Make sure the power supply for the TJA1104 is not controlled via INH when using JTAG because this pin will be switched off in the initialization of JTAG.

Please note that in case GPIO0, GPIO1, GPIO2 or GPIO9 are used for LEDs these will flicker while JTAG is active.

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4 Basic Control and Basic Status

4.1 Reset Options

A reset can be triggered by asserting RST_N while the device is powered up.

In addition, there are several reset options accessible via SMI as the TJA1104 offers to reset the whole device as well as individual submodules. The following table shows the reset options and the registers involved.

Table 28. Reset options

Module	Register	Description
Device Reset	DEVICE_CONTROL. DEVICE_ RESET (30.0040h.15)	Same as hardware or power on reset. All settings are lost and pinstrapping is executed
Soft Reset	DEVICE_CONFIG. DEVICE_ ENABLE (30.0048h.15)	Reasserting this bit triggers a soft reset. The following settings/registers are reset: PMA, PCS, INH_OVERRIDE
PMA Reset	PMA_CONTROL1. PMA_RESET (01. 0000h.15)	All PMA and PCS registers are reset. Internal PCS state reset. Resets symbol errors, link status drops, link available drops, link failures
PCS Reset	PCS_CONTROL1. RESET (03. 0000h.14)	Should only be used for debug purposes. Internal PCS state reset. PCS reset does not work in slave mode when polarity detection and correction is enabled, resulting in linkup not happening. *Workaround*: Use PMA reset instead of PCS reset.
PHY Reset	PHY_CONTROL. RESET (03.8100h. 15)	All PMA and PCS registers are reset. Resets symbol errors, link status drops, link available drops, link failures
PTP Reset	PTP_CONTROL. SOFT_RESET (30. 1101h.0)	Only PTP registers are reset.
PTP Reset	PORT_PTP_CONTROL. RESET (30. 9000h.15)	Only PTP registers are reset.
BIST Reset	PORT_BIST_CONTROL. RESET (30.A800h.15)	Bist settings, generator and checker stats are reset
BIST Status Reset	BIST_CHECK_CTRL. STATISTC_ CNT_RST (30.A8C0h.13)	Resets status: BIST_STATUS, GOOD_FRAME_CNT, BAD_ FRAME_CNT
GPIO & LED Reset	GLOBAL_INFRA_CONTROL. RESET (30.2C00h.15)	Resets GPIO and LED registers
CMOS xMII	PORT_INFRA_CONTROL. RESET (30.AC00h.15)	xMII related registers which are set by pinstrapping & OTP are set to what was latched during initial pinstrap sequence, e.g. XMII_DATA_CONFIG.
xMII block	XMII_CONTROL. RESET (30.AFC0h. 15)	Reloads OTP & pinstrapped configuration. xMII related registers which are set by pinstrapping are set to what was latched during initial pinstrap sequence, e.g. XMII_DATA_CONFIG.
SGMII block	SGMII_SR_MII_CTRL. RST (30. B000h.15)	Triggers a soft reset. All registers are set to their default values.
SGMII block	SGMII_VR_MII_DIG_CTRL1. VR_RST (30.B070h.15)	Triggers a soft reset. All registers are set to their default values.

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Table 28. Reset options*...continued*

Module	Register	Description
Port	PORT_CONTROL. RESET (30. 8040h.15)	The following settings/registers are reset: PMA, PCS, BIST generator/checker, PTP configuration, XMII configuration
xMII enable	PORT_FUNC_ENABLES. MII_ ENABLE (30.8048h.13)	Triggers a reset on 0 to 1 change All registers will be preserved
Bist enable	PORT_FUNC_ENABLES. BIST_ ENABLE (30.8048h.11)	Triggers a reset on 0 to 1 change Settings will be preserved Statistics will be reset
PTP enable	PORT_FUNC_ENABLES. PTP_ ENABLE (30.8048h.3)	Triggers a reset on 0 to 1 change
PHY enable	PORT_FUNC_ENABLES. EPHY_ ENABLE (30.8048h.2)	Triggers a reset on 0 to 1 change

Completion of device reset can be checked by reading the reset bit (30.0040.15) and observing the change to 0.

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4.2 PHY initialization

The PHY can be configured via pinstrapping to start up in autonomous or managed mode.

In autonomous mode no initialization is needed; the PHY is transitions automatically to normal mode and establishes a link.

In managed mode, the PHY is boots up in standby mode and needs to be configured manually. Figure 23 shows an example init flow. Detailed configurations for the submodule inititializations shown can be found in Section 5.

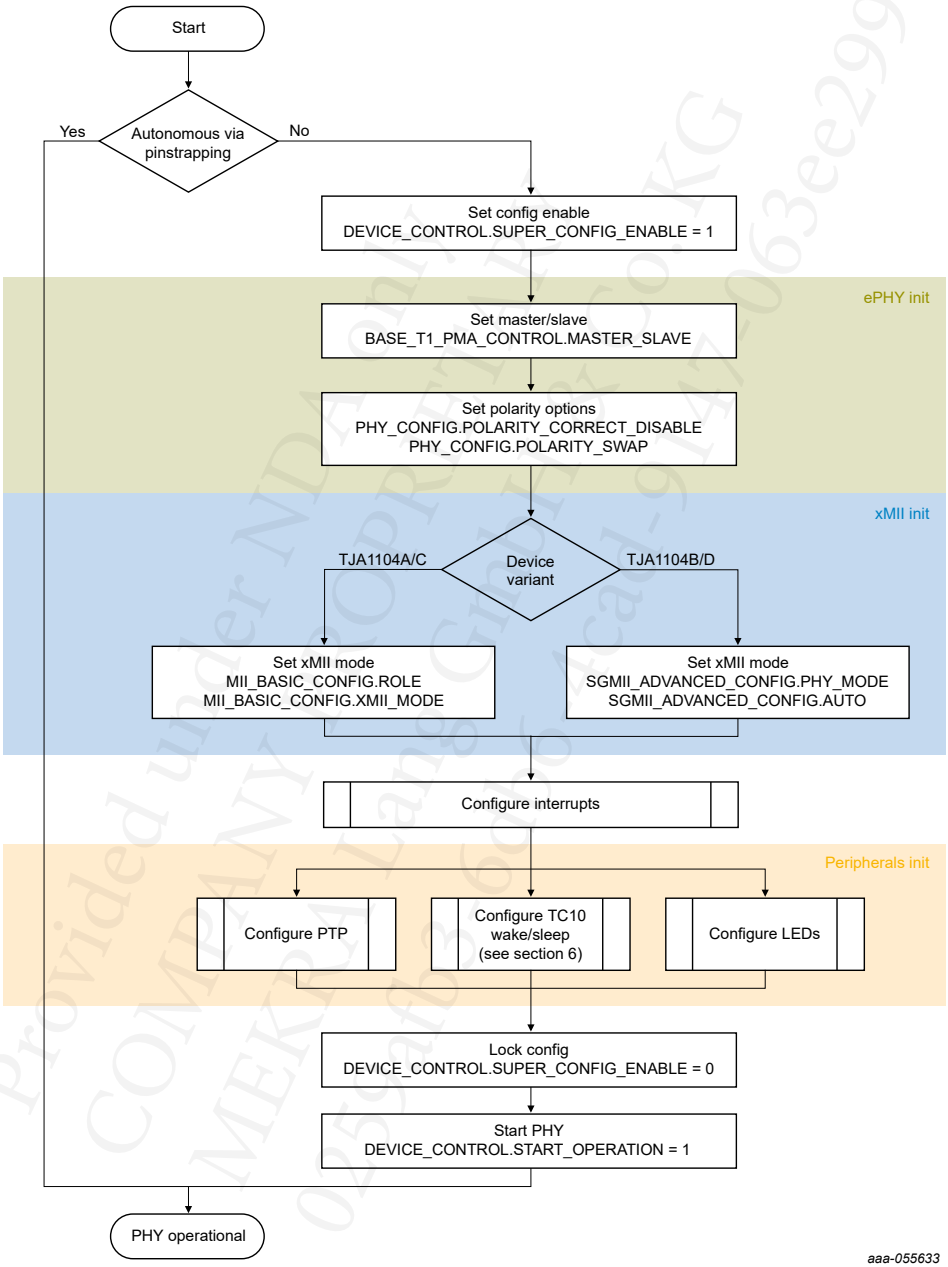


Figure 23. Example flow: complete PHY initialization

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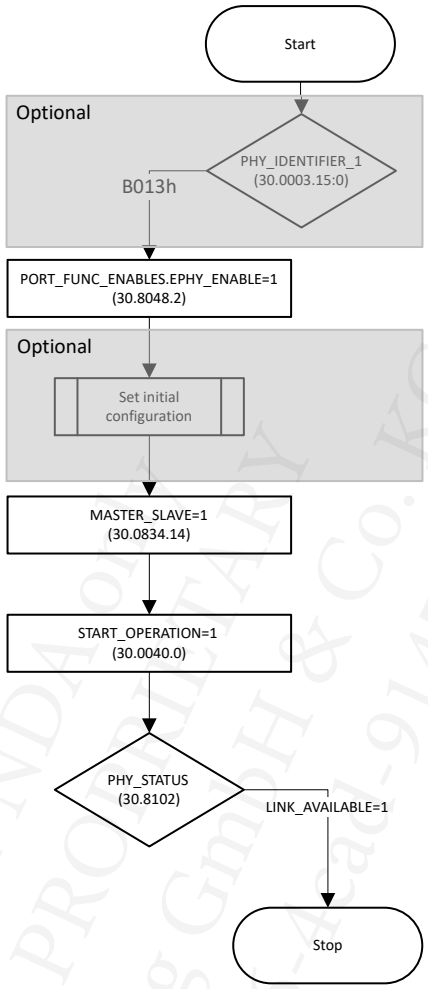


Figure 24. Example flow: Managed mode initialization

Note: *For TJA1104A/B, after power-on the MACsec block is configured in bypass mode and therefore all frames will be forwarded unencrypted and unsigned. If MACsec is used and it must be ensured that no transmission takes place before MACsec is configured the PHY must be pinstrapped to managed mode. That way no unintended linkup occurs.*

4.3 Configuration protection

Writing to the configuration registers is disabled by default to avoid unintended reconfiguration. To change the configuration, configuration register writing must be enabled. There are multiple, hierarchical CONFIG_ENABLE switches. Table 29 gives an overview which registers are protected in which way.

There is one device configuration enable bit (SUPER_CONFIG_ENABLE) which enables all reconfiguration. The configuration is expected to happen after power-on as part of the initialization routine; however, reconfiguration of the PHY is possible at any later time.

All other registers that are not listed in this table are not write reconfiguration protected.

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Table 29. Reconfiguration protected registers

Level 1	Level 2	Level 3	Protected registers
MMD30.DEVICE_CONTROL[SUPER_CONFIG_ENABLE]			
	MMD30.DEVICE_CONTROL[GLOBAL_CONFIG]		
			MMD30.DEVICE_CONFIG_EXTENDED
			MMD30.DEVICE_CONFIG
			MMD30.WISE_CONFIG
			MMD30.WISE_PARAMETERS
			MMD30.SMI_CONFIG
	MMD30.GLOBAL_INFRA_CONTROL[CONFIG_ENABLE]		
			MMD30.GPIO0_FUNC_CONFIG - MMD30.GPIO11_FUNC_CONFIG
	MMD30.PORT_CONTROL[CONFIG_ENABLE]		
		MMD30.PHY_CONTROL[CONFIG_ENABLE]	
			MMD1.BASE_T1_PMA_CONTROL
			MMD1.E100BT1_PMA_TEST_CONTROL
			SharedMMD.PHY_CONFIG
			SharedMMD.PHY_PARAMETERS
			SharedMMD.WAKE_SLEEP_CONFIG
			SharedMMD.WAKE_SLEEP_PARAMETERS
			SharedMMD.PHY_COMPLIANCE_TEST
		MMD30.PORT_BIST_CONTROL[CONFIG_ENABLE]	
			MMD30.BIST_INTERCEPT_CONFIG
		MMD30.PORT_INFRA_CONTROL[CONFIG_ENABLE]	
			MMD30.INFRA_CONFIG
			MMD30.MII_BASIC_CONFIG
			MMD30.XMII_CLK_CONFIG
			MMD30.XMII_CLK_IO_CONFIG
			MMD30.XMII_DATA_CONFIG
			MMD30.XMII_DATA_IO_CONFIG
			MMD30.RGMII_RXC_DELAY_CONFIG
			MMD30.RGMII_TXC_DELAY_CONFIG
			MMD30.RX_PREAMBLE_COUNT
			MMD30.TX_PREAMBLE_COUNT
			MMD30.RX_IPG_LENGTH RX IPG
			MMD30.TX_IPG_LENGTH TX IPG
		MMD30.SGMII_ADVANCED_CONTROL[CONFIG_ENABLE]	
			MMD30.SGMII_ADVANCED_CONFIG

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4.4 MASTER / SLAVE reconfiguration

The sequence for reconfiguring the MASTER/SLAVE mode is shown in Figure 25. As there are many overlapping configuration protection switches the sequence only shows one of many possible examples. Refer to Section 4.3 for more details.

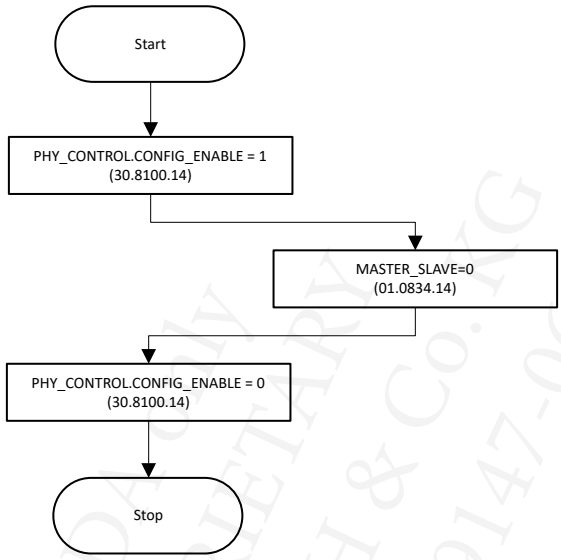


Figure 25. Example flow: Slave configuration

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4.5 xMII reconfiguration

The xMII mode can be changed via **MII_BASIC_CONFIG.XMII_SUB_TYPE** (30.AFC6h.2:0) at any time. The required sequence is listed in Figure 26.

Note: *Applies only to TJA1104A, for SGMII reconfiguration see Section 4.6*

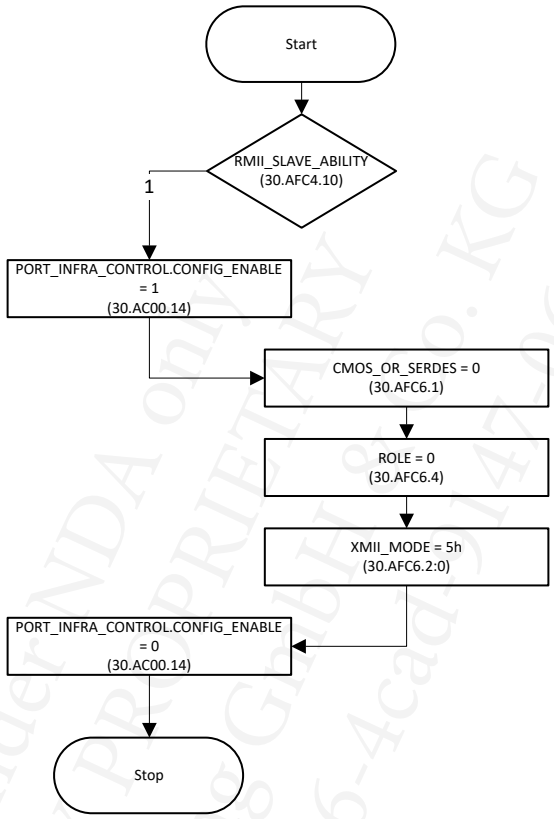


Figure 26. Example flow: xMII Mode change to RMII

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4.6 SGMII reconfiguration

The SGMII mode (PHY or MAC) can be configured via pinstrapping. Configuration via SMI allows overwriting the pinstrapping configuration as well as setting additional parameters.

Table 30 lists registers commonly used during SGMII configuration. For a full set of registers see register description in the datasheet starting MMD30 address B000h. (Section 7.4)

Table 31 lists registers shared between TJA1104A and TJA1104B which contain SGMII related fields.

Table 30. SGMII commonly used registers

Register	Description
MMD30.SGMII_SR_MII_CTRL 30.B000h	• Reset • Auto negotiation • Power down
MMD30.SGMII_SR_MII_STS 30.B001h	• Link status • Auto negotiation status
MMD30.SGMII_ADVANCED_CONTROL 30.B010h	• Config enable • Standby/Start operation
MMD30.SGMII_ADVANCED_STATUS 30.B011h	• Link status
MMD30.SGMII_ADVANCED_CONFIG 30.B013h	• PHY mode (PHY or MAC) • Operation mode (autonomous or managed)
MMD30.SGMII_IRQ_SOURCES 30.B014h MMD30.SGMII_IRQ_ENABLES 30.B015h MMD30.SGMII_IRQ_MSTATUS 30.B016h	Interrupt related registers • FIFO status • Autonegotiation status • Link status
SGMII_LED_TRIGGER0 30.B040h	SGMII LED status control
SGMII_VR_MII_DIG_CTRL2 30.B073h	Polarity inversion for Tx or Rx differential pair
SGMII_VR_MII_DIG_CTRL1 30.B072h	SGMII PCS loopback When SGMII PCS loopback is enabled, the first byte of the preamble is dropped. *Workaround*: none

Table 31. Shared registers containing SGMII related fields

Register	Description
PORT_FUNC_IRQ_ENABLES 30.807Ah PORT_FUNC_IRQ_STATUS 30.8078h PORT_FUNC_IRQ_MSTATUS 30.8074h	• Serdes IRQs
PORT_LEVEL_IRQ_ENABLES 30.8072h PORT_LEVEL_IRQ_STATUS 30.8070h PORT_LEVEL_IRQ_MSTATUS 30.8074h	• Serdes IRQs
FAVORITE_IRQS 30.8014h	• SGMII_LINK_EVENT
LEDx_TRIG_SOURCE 30.01A1h	• Port SGMII LED Trigger
XMII_ABILITIES 30.AFC4h	• SGMII_ABILITY
MII_BASIC_CONFIG 30.AFC6h	• CMOS_OR_SERDES • XMII_MODE

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Table 31. Shared registers containing SGMII related fields...*continued*

Register	Description
COMPOSITE_STATUS 30.8021h	• ALL_LINK_STATUS • LINK_STATUS_SELECT

4.7 Polarity Detection / Correction

The TJA1104 has polarity detection and configurable polarity correction. Polarity detection is always enabled, its status is reflected by the DETECTED_POLARITY bit. Polarity correction is enabled via pinstrapping or register settings. Additionally a forced polarity swap can be enabled, which can be used to debug polarity issues. The polarity correction is only available in SLAVE mode.

Note: *If no polarity correction is desired it can be disabled by setting PHY_CONFIG.DISABLE_POL_CORRECT (30.8108h.2) = 1, see Figure 27.*

Note that polarity detection is only performed once when the link is established. Enabling or disabling polarity detection has no effect without reconnecting the cable or restarting the process in another way. Furthermore POL_SWAP does invert polarity regardless of polarity detection. This has 2 non obvious effects:

1. Activating polarity swap on an established link will bring down the link, regardless of the polarity detection setting.
2. Activating polarity swap will invert the state of POLARITY_DETECTED flag the next time a polarity detection is performed.

Polarity options can be configured via the PHY_CONFIG (30.8108h) register.

Note:

Table 32. PHY_CONFIG polarity relevant part

Bit	Symbol	Value	Description
3	POLARITY_SWAP	0	No polarity swap
		1	Polarity swap enabled, TRX_P/TRX_N pins swapped
2	POLARITY_CORRECT_DISABLE	0	Polarity correction enable. Only possible in SLAVE mode
		1	No polarity correction

Figure 27 shows how to disable the polarity detection and enable polarity swapping. Both settings can be changed individually.

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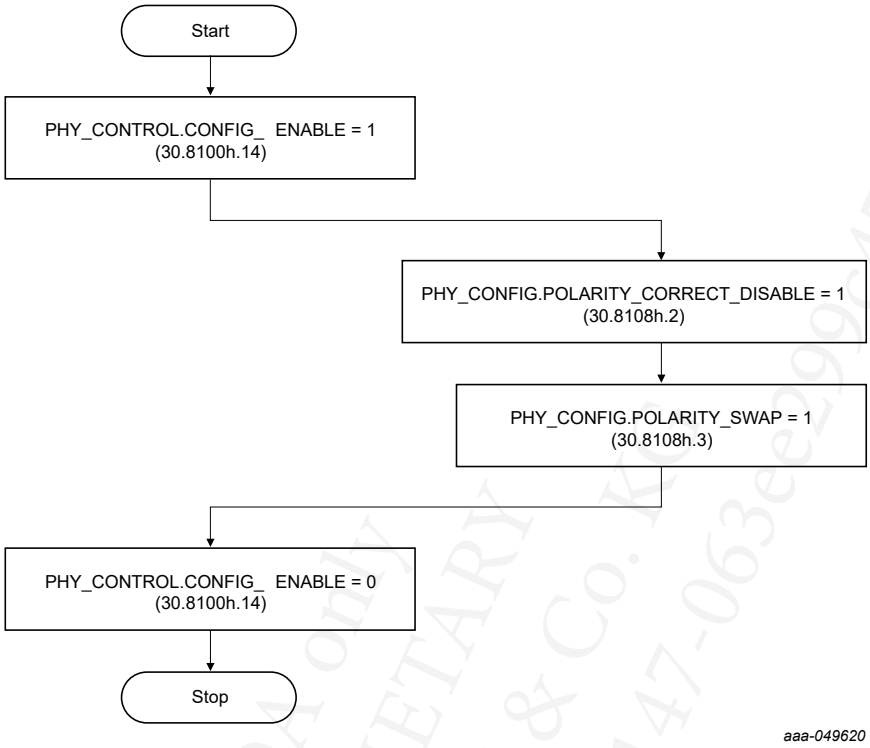


Figure 27. Example flow: Polarity settings

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4.8 Frequently used registers

The following registers might be worth examining at various situations.

Table 33. Registers of interest

State	Register
During link training	• MMD30.PHY_STATUS (30.8102h) – LINK_AVAILABLE (Bit 6)
On link up failure	• MMD30.PHY_STATUS (30.8102h) – SENDN_OR_DATA (Bit 5) – REM_RCVR_STAUS (Bit 4) – SCRAMBLER_STATUS (Bit 3) – LINK_STATUS (Bit 2) – LOC_RCVR_STATUS (Bit 1) • MMD30.PHY_STATE (30.810Ch) – BASIC_STATE (Bit 0:3) – TRANSMIT_STATE (Bit 8:11)
Polarity issues	• MMD30.PHY_STATUS (30.8102h) – DETECTED_POLARITY (Bit 7) • MMD30.PHY_CONFIG (30.8108h) – POLARITY_SWAP (Bit 3) – PLARITY_CORRECT_DISABLE (Bit 2)
SMI access issues	• ALWAYS_ACCESSIBLE (1Fh) – REFCLK_WARN (Bit 5) – FUSA_PASS_IRQ (Bit 4) – CORE_SUPPLY_STATUS (Bit 2) – LIMITED_ACCESS (Bit 1)
Diagnosis	• MMD1.PMA_CONTROL1 (01.0000h) – REMOTE_LOOPBACK (Bit 1) – LOCAL_LOOPBACK (Bit 0) • MMD3.PCS_CONTROL1 (03.0000h) – LOOPBACK (Bit 14) • MMD30.XMII_CONTROL (30.AFC0h) – MAC_LOOPBACK (Bit 1) – PHY_LOOPBACK (Bit 0) • MMD1.E100BT1_PMA_TEST_CONTROL (01.0836h) – PMA_TEST_MODE (Bit 15:13)
Master/Slave	• MMD1.BASE_T1_PMA_CONTROL (01.0834h) – MASTER_SLAVE (Bit 14)
MACsec (TJA1104A/B only)	Please refer to Section 5.6

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4.9 Operating modes

The following table provides information about the status of all pins during various operating modes. Transitions between modes are highlighted in the datasheet under section "6.1 Operating modes"

Table 34. Operating Modes

Pin	Type	Power off	Deep sleep	Standby Sleep Disable	Operational
Requirements					
VDD_CORE VREGA_IN VDDA_RX VDDA_TRX VDDIO		Off	Off	On	On
VDDA_AO		Off	On	On	On
Clock		Off	Off	On	On
RST_N		Don't care	High	High	High
Pin states					
MDC MDIO	Input/Output[1]	Unpowered	Unpowered	Active	Active
INT_N	Output[1] Low side open drain	Unpowered	Unpowered	Active	Active
XI XO		Unpowered	Unpowered	Active[2]	Active[2]
RST_N	Input[3] Internal pullup	Unpowered	Active	Active	Active
WAKE_IN_OUT	Input/Output[3] High side open drain	Unpowered	Floating[4]	Floating[4]	Floating[4]
INH	Output[3] High side open drain	Unpowered	Floating	High	High
TRX_P TRX_M	Analog[5]	Unpowered, high impedance	No bias, 100ohm differential termination	No bias, 100ohm differential termination	100ohm differential termination 0.9 V bias
GPIO0	Input/Output[1]	Unpowered	Unpowered	Output[4]	Output[4]
GPIO1 GPIO2 GPIO9	Input/Output[1]	Unpowered	Unpowered	Floating[4]	Floating[4]
RXD0-RXD3 RX_CTL RXC	Output[1]	Unpowered	Unpowered	Output, 50ohm	Output, 50ohm
TXD0-TXD3 TX_CTL	Input[1]	Unpowered	Unpowered	Input, high impedance	Input, high impedance

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Table 34. Operating Modes*...continued*

Pin	Type	Power off	Deep sleep	Standby Sleep Disable	Operational
TXC					

[1] Referenced to VDDIO, ESD diodes to GND and VDDIO present
[2] Depending on configuration either single ended input, differential input or crystal oscillator circuit
[3] Referenced to VDDA_AO, ESD diodes to GND and VDDA_AO present
[4] Power-on behavior, can be changed via configuration
[5] Referenced to VDDA_TRX, ESD diodes to GND and VDDA_TRX present

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5 Extended Control

5.1 Test Modes

TJA1104 supports four test modes, see Datasheet [1] for more details.

As an example, the following flow diagram shows the steps required to activate and deactivate test mode 5 (Power Spectrum Density). Other test modes can be activated in the same way.

Please make sure that the PHY is not in standby mode.

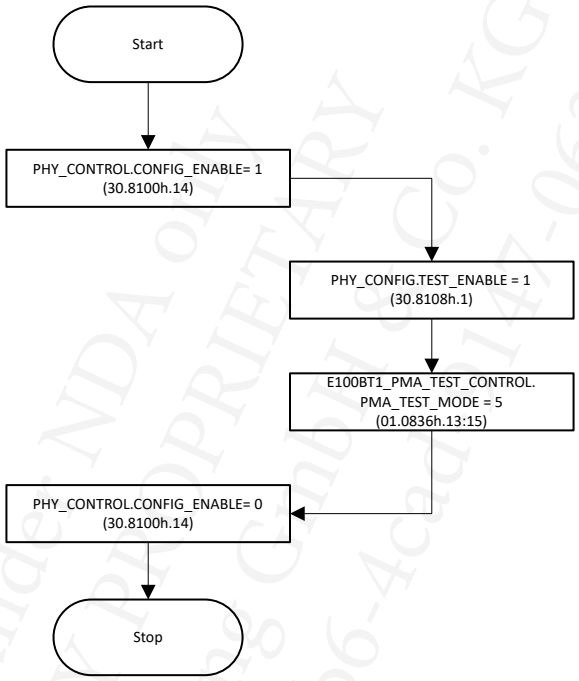


Figure 28. Example flow: entering test mode 5

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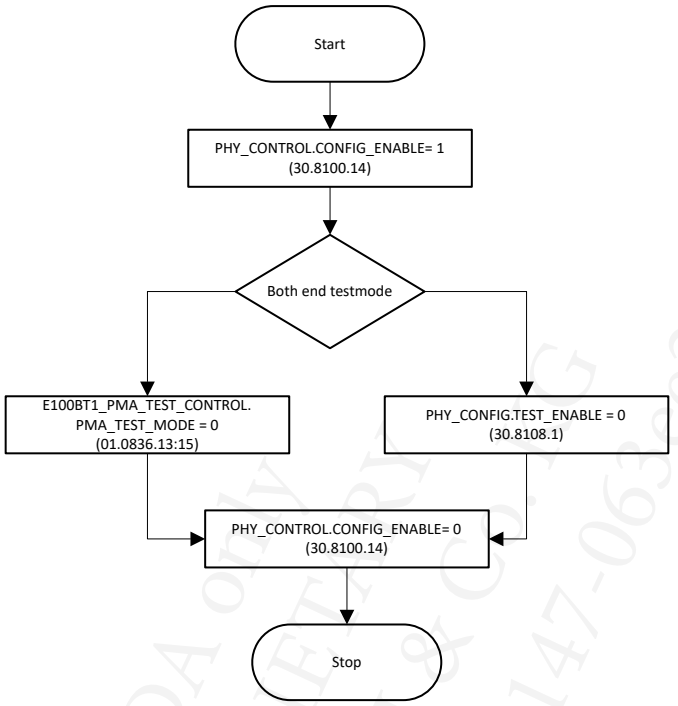


Figure 29. Example flow: exiting test modes

5.1.1 Recovered clock (TCLK)

The recovered clock can be made externally accessible on pin 26 (RXC/GPIO8/TX_TCLK). To enable TCLK, the following registers must be configured:

```
//Allow change of PHY_COMPLIANCE_TEST register
PHY_CONTROL.CONFIG_ENABLE (30.8100h.14) = 1

//Allow change of DEVICE_CONFIG register
DEVICE_CONTROL.GLOBAL_CONFIG (30.0040h.14) = 1

//Allow test modes
DEVICE_CONFIG.TEST_ENABLE (30.0048h.0) = 1

//Enable pin
PHY_COMPLIANCE_TEST.TX_TCLK_EN (30.83F0h.0) = 1

// Activate the test interface
// If executed as one register write this is 30.03F4h = 8120h
COMPL_TEST_INTF.TEST_INTF_EN (30.03F4h.15) = 1
COMPL_TEST_INTF.FUNC_SEL (30.03F4h.11:8) = b0001
COMPL_TEST_INTF.COMPL_TCLK_EN (30.03F4h.5) = 1
```

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5.2 Cable test

A cable test checks for shorted or open bus cables. In general there must be no activity on the MDI interface while the cable test is in progress. For this reason it is recommended to perform the cable test only when configured as master (because this implies the link partner is slave). Alternatively it must be ensured that a link partner in master mode is disabled before starting the test. Otherwise the measurement yields wrong results. The cable test can be triggered by enabling and starting the cable test as follows.

```
CABLE_TEST.ENABLE (30.8330h.15) = 1
CABLE_TEST.START (30.8330h.14) = 1
```

After the cable test is finished, the valid bit CABLE_TEST.VALID (30.8330h.13) is set. The result can be retrieved via CABLE_TEST.FAULT_TYPE (30.8330h.2:0). The following conditions are detected:

Condition	Details
No cable fault detected	Occurs when the cable is terminated correctly, e.g. a link partner is connected
Shorted pair	
Open pair	
Unable to detect	Occurs for example when a link partner is talking or a high noise level is present on the cable

5.3 Loopback

Loopback operation can be used for diagnosis purposes, especially to locate potential error sources. Ethernet frames are transmitted by the MAC and looped back at some point of the data path. If the loop is error-free, the MAC receives the same frames as sent out before.

In accordance with the 100BASE-T1 IEEE specifications, the TJA1104 provides the following loopback modes: Internal, external, xMII, and remote loopback.

In case of internal loopback, the loop point is after the PCS block. It checks proper operation of the internal PCS part. In case of external loopback, the loop point is after the PMA block. It checks proper operation of both the internal PCS and PMA part. The remote loopback spans the entire loop from one PHY to the other PHY and back to the first PHY. It checks the entire channel.

Table 35 gives information about activating different loopback modes.

Table 35. Loopback modes

Mode	Register	Precondition	Additional Info
Internal (PCS)	PCS_CONTROL1. LOOPBACK (03.0000h.14)		Link available always set
External (PMA)	PMA_CONTROL1. LOCAL_LOOPBACK (01.0000h.0)	• cable must be open, not terminated or connected to link partner	Link available set when cable is open
Remote (PMA)	PMA_CONTROL1. REMOTE_LOOPBACK (01.0000h.1)	• Link partner connected	Link available set when link partner connected
xMII PHY	XMII_CONTROL. PHY_LOOPBACK (30.AFC0h.0)	• INFRA_CONFIG. TEST_ENABLE set • Link partner connected	Link available set when link partner connected
xMII MAC	XMII_CONTROL. MAC_LOOPBACK (30.AFC0h.1)	• INFRA_CONFIG. TEST_ENABLE set	Link available set when link partner connected

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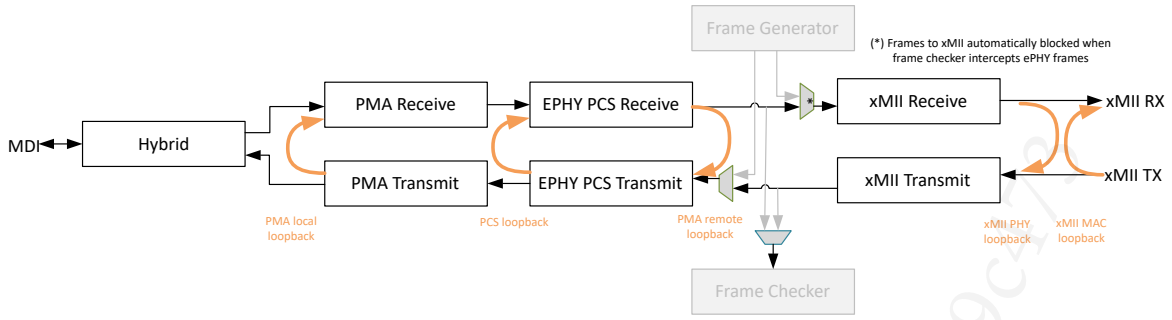


Figure 30. Supported loopbacks

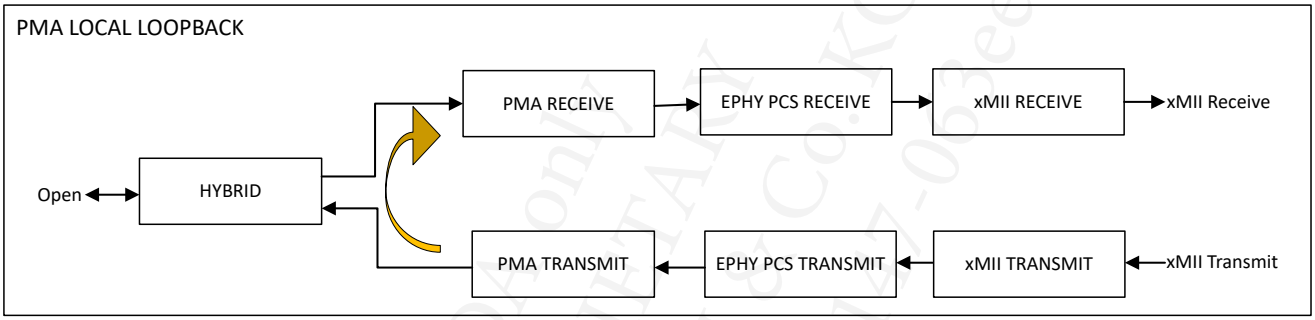


Figure 31. External (PMA local) loopback

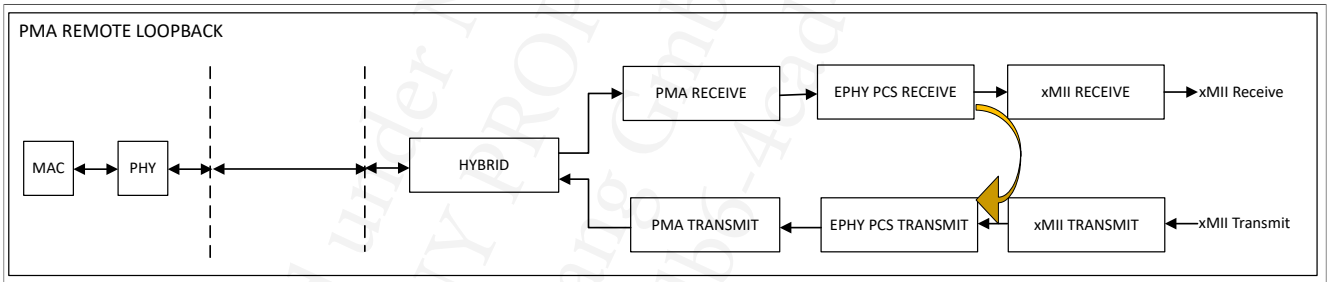


Figure 32. Remote (PMA remote) loopback

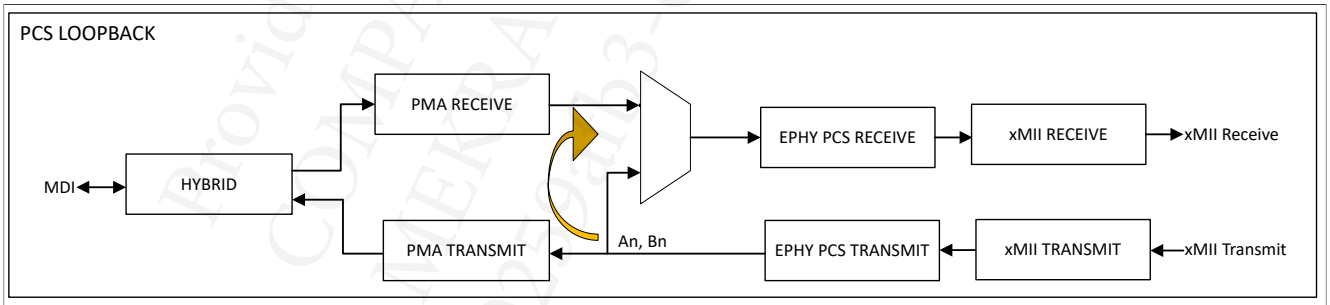


Figure 33. Internal (PCS) loopback

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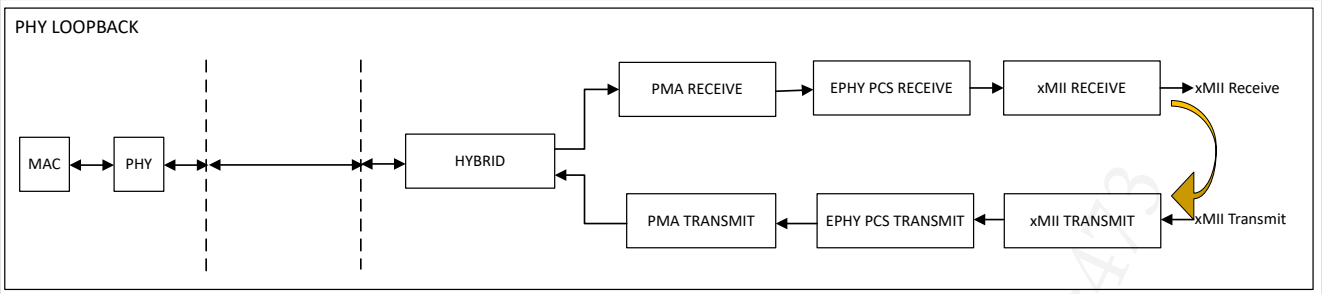


Figure 34. xMII PHY loopback

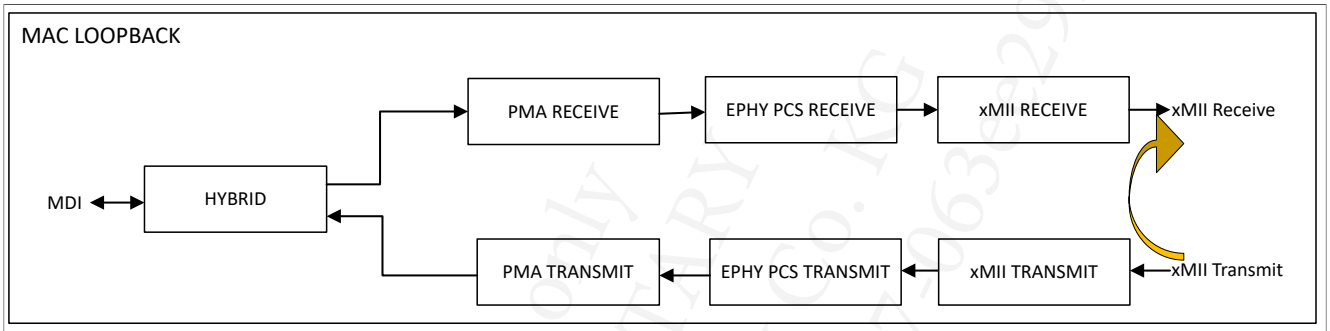


Figure 35. xMII MAC loopback

5.3.1 xMII Dependencies

Some combinations of loopback mode and xMII mode do have requirements on the xMII pads although data is looped back before reaching the xMII pads. For initial bring up test this must either be taken into account or a xMII modes that doesn't have such a dependency must be chosen. Table 36 shows pads which must either not be disturbed or have a specific function associated with when activating loopback modes.

Table 36. Loopback pad dependencies

	MII	MII rev	RMII	RMII clk	RMII rev	RGMII
xMII PHY Loopback	RXC	RXC	TXC	TXC	TXC	RXC
Remote (PMA) Loopback	RXC TXC			TXC		RXC

Loopback modes which utilize xMII pads as part of its datapath are not shown since xMII dependencies are obvious.

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5.4 BIST Generator/Checker

The BIST (built-in self-test) is an integrated Ethernet frame generator and checker intended for diagnosis. Both generator and checker can be enabled independently. The BIST generator provides full configurability on Ethernet frame header with programmable inter-packet gap (IPG). The BIST checker collects statistics on received frames.

5.4.1 Generator

The frame generator can operate in 2 different modes: Continuous or production.

In continuous-mode, an infinite stream of good frames according to the frame format settings are sent. Good frames are frames that have a valid CRC code and are transmitted without RX_ER asserted. The generator will only stop sending frames after BIST_GEN_CTRL.STOP is set (30.A880h.12).

In production-mode, the frame format and the number of good and bad frames to be transmitted are configured. On completion, the generator stops automatically and the BIST_GEN_STATUS.BIST_GEN_DONE flag is set (30.A881h.15). In most usecases the generator production-mode will be combined with checker production-mode in which case the starting of both modules needs to be synchronized. Please have a look at Section 5.4.2 for more details.

The following frame properties can be changed. Please have a look at the datasheet for specific registers/addresses

- Preamble length (fixed, incrementing, random)
- Source MAC address
- Destination MAC address
- Source MAC address
- Ethernet type
- Payload data (1 byte fixed, 1 byte incrementing, random)
- Payload length (fixed, incrementing, random)
- Ratio of PTP messages, type of PTP message

5.4.1.1 Example

The power-on default values of all BIST generator related registers are usable defaults which allows testing the module with minimal register writes.

The following sequence gives a good starting point for BIST configuration for two reasons:

- The datapath uses an internal loopback mode eliminating any dependency on linkpartner or MAC
- Usage of power-on defaults where possible minimizes configuration complexity

```
//enable PHY + BIST block
30.0040h = 0x2001 //MMD30.DEVICE_CONTROL.SUPER_CONFIG_ENABLE = 1
                  //MMD30.DEVICE_CONTROL.START_OPERATION = 1
30.A800h = 0x4000 //MMD30.PORT_BIST_CONTROL.CONFIG_ENABLE = 1
30.8048h = 0x2805 //MMD30.PORT_FUNC_ENABLES.BIST_ENABLE=1
                  //MMD30.PORT_FUNC_ENABLES.TEST_ENABLE=1

//enable PCS loopback
03.0000h = 0x6000 //MMD3.PCS_CONTROL1.LOOPBACK = 1

//enable datapath BIST->MDI->PCS loopback->MDI->BIST
30.A807h = 0x0820 //MMD30.BIST_INTERCEPT_CONFIG.TX_INTERCEPT=2
                  //MMD30.BIST_INTERCEPT_CONFIG.CHECK_EPHY_OR_XMII=1

//enable BIST checker
30.A8C0h = 0x4001 //MMD30.BIST_CHECK_CTRL.BIST_CHECK_MODE=1
                  //MMD30.BIST_CHECK_CTRL.BIST_CHECKER_EN=1
```


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```
//enable BIST generator
30.A880h = 0x4000 //MD30.BIST_GEN_CTRL.BIST_GEN_EN = 1

//Wait 1 second

//Read frame counter:
30.A8D0h = ? //MMD30.R_GOOD_FRAME_CNT
```

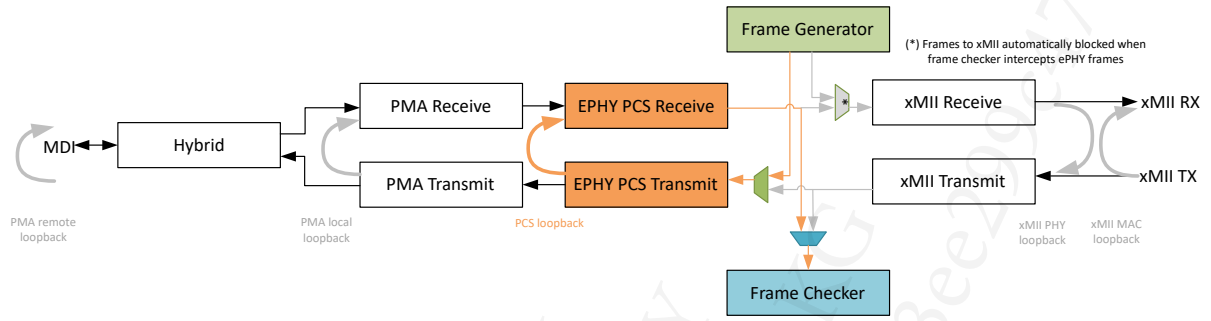


Figure 36. Active datapath in example

5.4.2 Checker

Similar to the generator, the checker can operate in continuous and production-mode.

In continuous-mode, received frames are analyzed and statistics for good, bad and RX_ER frames are generated. The statistics can be accessed via three registers. The continuous mode is running indefinitely and must be stopped by deasserting BIST_CHECKER_EN.

In production-mode, the number of received frames is compared to the number of frames configured in the generator. The checker only accepts frames in a time window which starts when asserting BIST_CHECKER_EN and ends after BIST_WAIT_TIMER microseconds. After the timer expired the flag BIST_PROD_STATUS.BIST_CHECK_DONE is asserted and the number of frames configured for the generator is compared with the number of frames received by the checker. BIST_PROD_STATUS.BIST_CHECK_FAIL is asserted in case the values do not match.

The counter for good and bad frames (R_GOOD_FRAME_CNT/R_BAD_FRAME_CNT) can be cleared by setting BIST_CHECK_CTRL.STATISTIC_CNT_RESET. The R_RXER_FRAME_CNT register is only cleared by disabling the BIST block.

Note: *Frame counters in the bist module do not saturate but wrap around after reaching 65535. By ensuring a maximum of one wrap arounds between read outs the actual frame number can still be calculated.*

Note: *When using the BIST checker in production-mode neither the state of BIST_PROD_STATUS nor the content of R_GOOD_FRAME_CNT/R_BAD_FRAME_CNT is frozen on BIST_WAIT_TIMER expiry. This may cause confusion for the following reasons:*

- *BIST_PROD_STATUS.BIST_CHECK_FAIL evaluation stays active and is not limited to the time set by BIST_WAIT_TIMER.*
- *Counters do not necessarily reflect the status the BIST_PROD_STATUS decision was based on as it is not possible to read both atomically.*

A minimal example showing the BIST checker function can be found in Section 5.4.1.1

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5.4.3 Datapath

The frame checker as well as the frame generator can each be configured to connect to either ePHY or MII. Figure 37 shows the datapath and relevant multiplexers involved.

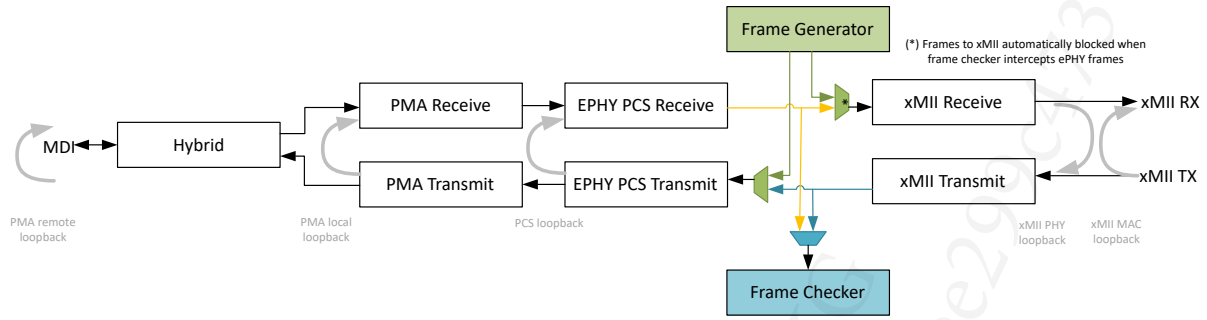


Figure 37. Datapath for frame generator/checker

The multiplexers can be configured to support the following modes:

Table 37. Datapath list

Generator direction	Checker direction	Usecase	Comment
Towards xMII	From xMII	• Mode 1 (PHY loopback) • Mode 2 (external (MAC) loopback)	ePHY disconnected
Towards ePHY	From ePHY	• Mode 3 (PMA local loopback) • Mode 4 (external (linkparter) loopback)	xMII disconnected
Towards xMII	From ePHY		Datapath from xMII to ePHY enabled
Towards ePHY	From xMII		Datapath from ePHY to xMII disabled

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5.5 Timestamping

The TJA1104 contains a IEEE 1588 v2 (PTP) module which allows timestamping inbound and outbound frames with very high precision. The following diagram shows the main function blocks participating in the ptp operation.

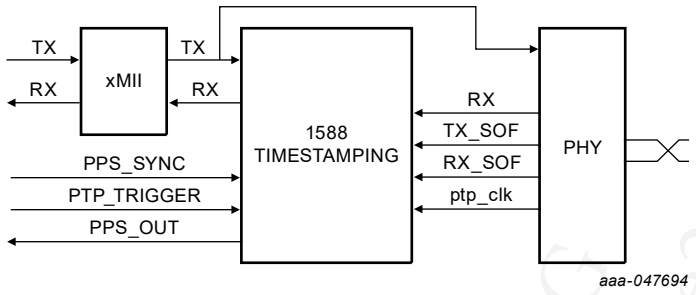


Figure 38. PTP function blocks

In order to get high precision timestamps the internal local time counter (LTC) must either be rate corrected to an externally provided reference clock, pose as the reference clock itself or ensure synchronized operation by other means.

- The rate correction can be achieved by 2 methods:
  - Hardware self-locking clock (1pps PLL)
  - Software assisted rate correction
- Using the LTC as reference for external devices can be achieved by making use of the PPS_OUT signal.

5.5.1 Hardware self-locking clock

Using the hardware self-locking clock requires the PPS_SYNC to be connected to a reference clock with known frequency and known absolute timestaps for every edge of the signal. The operation of the hardware self-locking clock works in 3 steps:

1. Basic initialization, e.g. enabling PTP and setting PTP_CLOCK_PERIOD
2. Initializing the LTC with a correct timestamp
3. locking the LTC to a reference signal

The initialization of the LTC works by first setting the interval of the PPS_SYNC signal in the HW_EXT_PPS_NS_x registers (0 s - 1.073 s in 1ns steps). The duty cycle does not matter as only the rising edge is used for synchronization. As a safeguard, the register HW_LOCK_ER_LMT_x can be used to set the maximum allowed deviation between the external PPS_SYNC and the internal nominal LTC. (0 ms - 16 ms in 1 ns steps) An interrupt is asserted if the limit is exceeded.

The next step is to load the timestamp of any upcoming rising edges of the PPS_SYNC signal into the LTC_WR_NSEC_x / LTC_WR_SEC_x registers. Finally the LOAD_LTC_PPS bit in the LTC_LOAD_CTRL register must be set at an arbitrary time in the clock cycle preceeding the choosen rising edge. The timestamp will be loaded into the LTC when the next rising edge of the PPS_SYNC signal occurs.

The locking of the LTC is done automatically.

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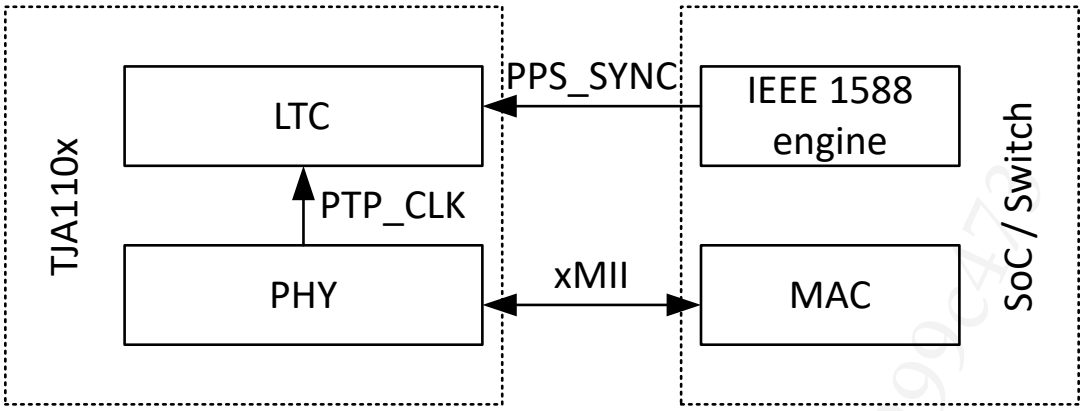


Figure 39. PTP hardware signals

5.5.2 Software assisted rate correction

Alternatively to the hardware self-locking clock a software assisted rate correction can be used. In order to match rate differences as well as correct offsets a constant rate correction and a temporary rate correction is required. The details of those concepts are described in the datasheet [1] under section 6.11.5.2 / 6.11.5.3.

To be able to calculate these rate correction values the ltc value and reference timestamp of some event must be known. This can be achieved by using the event timestamping mechanism which requires connecting the PTP_TRIGGER signal to the host controller.

5.6 MACsec

The TJA1104A/B includes a MACsec block. This block implements Ethernet frame encryption and authentication as described in the IEEE802.1AE standard.

The MACsec functionality is split into two sub modules:

1. The *MACsec Adapter*, which connects the RX and TX data path to the MACsec IP.
2. The *MACsec IP*, which implements the actual MACsec functionality.

Both MACsec sub modules can be configured using the SMI interface and are mapped into the MMD31 region. However, all MACsec registers are 32-bit wide, while SMI only offers 16-bit wide access. The next section introduces the register access to the MACsec block.

5.6.1 Register access

All MACsec registers are mapped into MMD region 31 as shown in Table 38. The MACsec registers can be found starting from SMI address 9000h up to address 9FFFh. The registers for the MACsec Adapter and the MACsec IP can be found inside this region. Three points need to be considered, when accessing these registers:

1. All MACsec registers as they are described in the datasheet are 32bit wide. However, SMI only allows 16bit access.
2. All MACsec register addresses in the datasheet are given as byte address. This results in an increment (of the byte address) of four between two neighboring MACsec registers. Accordingly, the SMI address difference between two neighboring MACsec registers is two.
3. All MACsec register addresses are given as relative byte address offset to the base address of each sub module (MACsec Adapter or MACsec IP). See table Table 39.

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Table 38 shows how the different address spaces are mapped into each other. Additionally it also shows how to access a complete 32bit MACsec register, with two consecutive SMI accesses. The lower half (Bit 0…15) of each 32bit MACsec register can be accessed with an even SMI address, while the upper half (Bit 16…31) is accessed with an odd SMI address.

Table 38. 32-bit MACsec registers in MMD31

MACsec register (32bit word)			SMI Address (16bit)	MACsec register byte address
reserved				
MACsec	Word 2047	Upper	9FFFh	1FFCh
		Lower	9FFEh	
	⋮			
	Word N	Upper	9001h + N × 2	N × 4h
		Lower	9000h + N × 2	
	⋮			
	Word 1	Upper	9003h	0004h
		Lower	9002h	
	Word 0	Upper	9001h	0000h
		Lower	9000h	
reserved				

The registers addresses for each MACsec sub module are given in the datasheet as byte offsets relative to each MACsec sub module. Table 39 shows the offset for each MACsec sub module.

Table 39. Address offset of MACsec sub modules

Name	Relative sub module offset (in bytes)	Absolute SMI offset
MACsec Adapter	0F00h	9780h
MACsec IP	0000h	9000h

The SMI address of the lower half word can now be calculated using the MACsec sub module offset (in bytes) and the register offset inside the MACsec sub module (in bytes):

$$\text{SMI address} = 9000\text{h} + \frac{\text{sub module offset} + \text{register offset}}{2}$$

The upper half word can be accessed at the subsequent SMI address.

Example: The ADAPTER_STATUS register in the MACsec Adapter is at offset 004h. The resulting SMI address of the lower half word is:

$$9000\text{h} + \frac{0\text{F00h} + 004\text{h}}{2} = 9782\text{h}$$

Consequently, the address of the upper half word in then 9783h.

The following pseudo code shows a function to read a 32bit MACsec register:

```
function macsec_read(sub_module_offset, register_offset):
    smi_address_lower = 9000h + (sub_module_offset, register_offset)/2

    lower = smi_read(MMD31, smi_address_lower)
    upper = smi_read(MMD31, smi_address_lower + 1)
```


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```
    return (upper << 16) + lower
```

Note, that for reading registers a SMI access with address auto-increment can also be used.

5.6.2 Example configuration

This example demonstrates how to configure the MACsec subsystem to send confidentiality and or integrity protected Ethernet frames. For this, both the MACsec Adapter as well as the MACsec IP need to be configured. The MACsec Adapter is enabled as following:

- Set ADPTR_ENA bit in ADPTR_CNTRL

The MACsec IP is configured next:

- Select a Secure Channel of the MACsec IP TX_SC_ID = 0…31
  - Note: this ID is used to select the internal registers of the MACsec IP, it is not the Secure Channel Identifier used in the SecTAG of the Ethernet frame
- Enable access to the SAK and SAH registers by writing the TX_SC_ID to TX SC access select (MACSEC_TXSCKA)
  - Write a key in the Security Association A via:
    - TX Security Association A key 1 of 8 (MACSEC_TXSAKA1Q)
    - TX Security Association A key 2 of 8 (MACSEC_TXSAKA2Q)
    - and so on
- Enable access to the Secure Channel configuration registers by writing the TX_SC_ID to TX SC access select (MACSEC_TXSCA)
  - Initialize the TX Security Association A nextPN (MACSEC_TXSAANPN) to 1
  - Initialize the TX Security Association Extended A nextPN (MACSEC_TXSAAXNPN) to 0
  - Activate the newly configured Security Association by clearing the `A` bit in TX Security Association A configuration and status (MACSEC_TXSAACS)
  - Set the Secure Channel Identifier to be used in the SecTAG
    - TX Secure Channel Identifier first half (MACSEC_TXSCI1H)
    - TX Secure Channel Identifier second half (MACSEC_TXSCI2H)
  - Configure the SC TX Secure Channel configuration (MACSEC_TXSCCFG):
    - Select the used cipher with CS
    - Configure CE to for confidentiality protection
    - Set AIS to always send the SCI
    - Configure (VLAN)-Tag handling with TCM and TVM as needed
    - Set PF to enable protection
- Configure the TX Secure Channel Selection, which classifies TX frames to a Secure Channel, for example based on source MAC address:
  - Write the source MAC address (i.e., the address of the MAC the TJA1104 is connected to) to MAC_SA in TX SC Selection Record 0 part 0-3 (MACSEC_TXSCSR0Px)
  - In TX SC Selection Record 0 part 3 (MACSEC_TXSCSR0P3)
    - Set USA to enable source MAC address matching
    - Set EN to enable this record
    - Write your TX_SC_ID into the TX_SC field
  - With this configuration, all matching frames are now selected to be sent using the previously configured Secure Channel
- Disable Bypass Mode by setting BYPN in MACsec configuration (MACSEC_CFG)

The MACsec Adapter and MACsec IP are now configured to transmit frames.

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5.6.3 Additional MACsec register information

5.6.3.1 Wrong SMI data on Clause 45 MACsec register read

When accessing registers in the MACsec register range via Clause 45 SMI read, the returned data may be randomly identical to the previously returned data and therefore wrong.

Workaround: Implement a software workaround to ensure the read data is correct.

Proposed sequence:

```
//only execute once after start up: initialize read_reg to known value
count = 0
do
last_read = <Read_Register TAG_PARAM>
while (last_read ≠ 0x88A8_8100) AND (count < 5)
//execute whenever a register is read
count = 0
do
//read actual register of interest
current_read = <Read_Register>
count = count + 1
//repeat when same data has been read assuming the read failed
//stop after 5 iterations in case read was successful and is
//in fact equal to last_read
while (current_read == last_read) AND (count < 5)
last_read = current_read
//current_read contains desired register value
```

5.6.3.2 SMI register read of MACsec registers fails

SMI register read of MACsec registers fails under certain conditions. When reading MACsec registers while MACsec is disabled, subsequent reads after enabling MACsec will return the wrong data. This includes waking up from sleep, which automatically enables MACsec if it was enabled before going into sleep.

Workaround: Implement a software workaround to ensure correct operation after enabling MACsec and wake up from sleep.

Proposed sequence:

```
//Enable MACsec
Set PORT_FUNC_ENABLES[MACSEC_ENABLE] (address Shared.8048h.5)
//Read any MACsec core register (e.g. revision)
Read MACSEC_IP_REV_1 (address MMD31.9000h)
//Disable MACsec
Clear PORT_FUNC_ENABLES[MACSEC_ENABLE] (address Shared.8048h.5)
//Enable MACsec
Set PORT_FUNC_ENABLES[MACSEC_ENABLE] (address Shared.8048h.5)
```

5.6.3.3 Invalid FCS when setting MACSEC BYPN

When changing MACSEC_CFG.BYPN back to 1, the first frame has an invalid FCS.

Workaround: None, don't disable/bypass MACsec during active communication.

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5.6.3.4 MACsec VLAN ID bypass issue when VCT is disabled

Frames received with a bypassed VLAN ID are not forwarded by the transceiver. The transmit direction is not affected.

Workaround:

1. Set VCT to 1 to avoid dropping frames in the transceiver. Refer to data sheet section 6.13.1.2 (MACsec configuration).
2. If a VLAN tag is expected after a secure tag, TVM and TCM must be set to 0. Refer to data sheet section 6.13.1.63 (TX secure channel)

5.6.3.5 Register read operations using CK45 transactions not reaching the MACsec IP

In the after port reset test, register read for the MACsec is performed using CL45 transactions. Read requests may not reach the MACsec IP.

Software workaround:

1. Initialize last_read_data_reg to '0xFFFF_FFFF'
2. All MACsec IP reads must be called in a loop
   - Exit loop when read value doesn't match the 'last_read_data_reg'
     # Accept the read data
     # Update last_read_data_reg with new read data
     # Increment count
   - Exit loop on count of 5 if read data always matches the stored value
     # Accept the read data
     # Update last_read_data_reg with new read data

Workaround added to the driver. For more details, see https://jira.sw.nxp.com/browse/AASW-12906 or https://bitbucket.sw.nxp.com/projects/ARTD/repos/ethtrcv/pull-requests/525/overview

5.7 Temperature warning limit

The device monitors contain a temperature warning limit. As soon as the predefined threshold is exceeded MMD30.TEMP_STATUS.HIGH_TEMP_NOW is asserted and an interrupt is generated. There is no automatic shutdown triggered. The host controller needs to monitor the respective bit and disable the PHY on overtemperature in order to protect the PHY from damage.

The temperature warning threshold can be adjusted via MMD30.TEMP_STATUS.TEMP_LIMIT. The formula for calculating the register value corresponding to a given temperature is:

```
T = 7.22*(TEMP_LIMIT/100)^2 + 38.8*(TEMP_LIMIT/100) + 65.3
```

where T is the temperature in degree celsius and TEMP_LIMIT is the value specified in the register. Please note that due to the temperature range supported by the sensor only register values between 41h and BFh (93°C - 165°C) correspond to valid limit temperatures.

The following linear approximation results in less than 1.5°C error between 93°C and 165°C.

```
T = TEMP_LIMIT*1.75 - 96.5
```

The preconfigured register value is A1h which corresponds to 146°C. Table 40 lists relevant register/temperature pairs.

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Table 40. Temperature limit values

Register	Temperature	Comment
0h - 40h	65°C - 93°C	Temperature range not covered by the sensor
41h	93.5°C	Lowest temperature selectable
4Fh	100°C	
A1h	146°C	Predefined default
A7h	150°C	
BFh	165.7°C	Highest temperature selectable
C0h - FFh	166°C - 211°C	Temperature range not covered by the sensor

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6 Power Modes

6.1 Wake-up/Sleep (TC10)

The following sections demonstrate how to use TC10 wake-up/sleep capabilities.

6.1.1 Hardware considerations

- Always-on power supply (Pin VDDA_AO)
  - VDDA_AO must be kept supplied during sleep to remain wake-able
- Wake-up signal (Pin WAKE_IN_OUT)
  - This pin allows for local wake-up (input) or wake forwarding (output)
  - The output stage of this pin is an high-side switch which allows connecting several wake-up sources to the same signal (wired or-ing)
  - When configuring this pin as input and output simultaneously, pulse elongation (e.g. due to capacitances) must be avoided to prevent wake forward loopbacks
  - Notice WAKE_IN_OUT is in the VDDA_AO domain
- Inhibit signal (Pin INH)
  - This signal can be used to control external power supply
  - The output stage of this pin is an high-side switch which allows connecting several inhibit sources to the same signal (wired or-ing)
  - At startup, as soon as VDDA_AO is applied the INH signal is pulled high
  - If the TJA1104 is in sleep mode, it switches off the INH signal which indicates the supply can be disabled
  - To avoid unwanted supply shutdown, the WISE_CONTROL.INH_OVERRIDE bit is set after every reset and wake-up. In order to release INH it must be cleared before transitioning to sleep mode.
  - If INH is released, the always-on domain (i.e. the 100BASE-T1 subblock) gets isolated.
  - A wakeup is needed to release the isolation, this can be WAKE_IN_OUT, remote wakeup or ALWAYS_ACCESSIBLE.WU_SMI if VDDIO is available.

6.1.2 Configuration

The wake-up/sleep concept is designed to support multi-port PHY devices and differentiates between device configuration (WISE_... registers) and port configuration (WAKE_SLEEP_... registers). As this is a one-port device the port configuration block is present once. This concept has several implications:

- Wake pin configurations is global, but the reaction (for every port) is defined separately
- Wake-up/sleep functionality can be enabled/disabled globally
- Wake-up/sleep featureset (TC10 / Basic sleep) can be selected per port

The following diagram shows a typical TC10 wake-up/sleep configuration. It is assumed all registers initially contain their power-on values. Except for the initial write to CONFIG_ENABLE the order of configuration does not matter. WISE_STATUS.DEV_SLEEP_STATUS and WISE_STATUS.WU_IO_RECEIVED mirror the states of WAKE_SLEEP_STATUS.SLEEP_STATUS and WAKE_SLEEP_STATUS.WU_IO_RECEIVED on TJA1104, therefore only one needs to be checked. Interrupts can be enabled by setting TOP_EPHY_IRQ_ENABLES. WAKE_SLEEP_EVENT=1 (global IRQs) or EPHY_FUNCTIONAL_IRQ_ENABLES.WAKE_SLEEP_EVENT=1 (per port IRQs)

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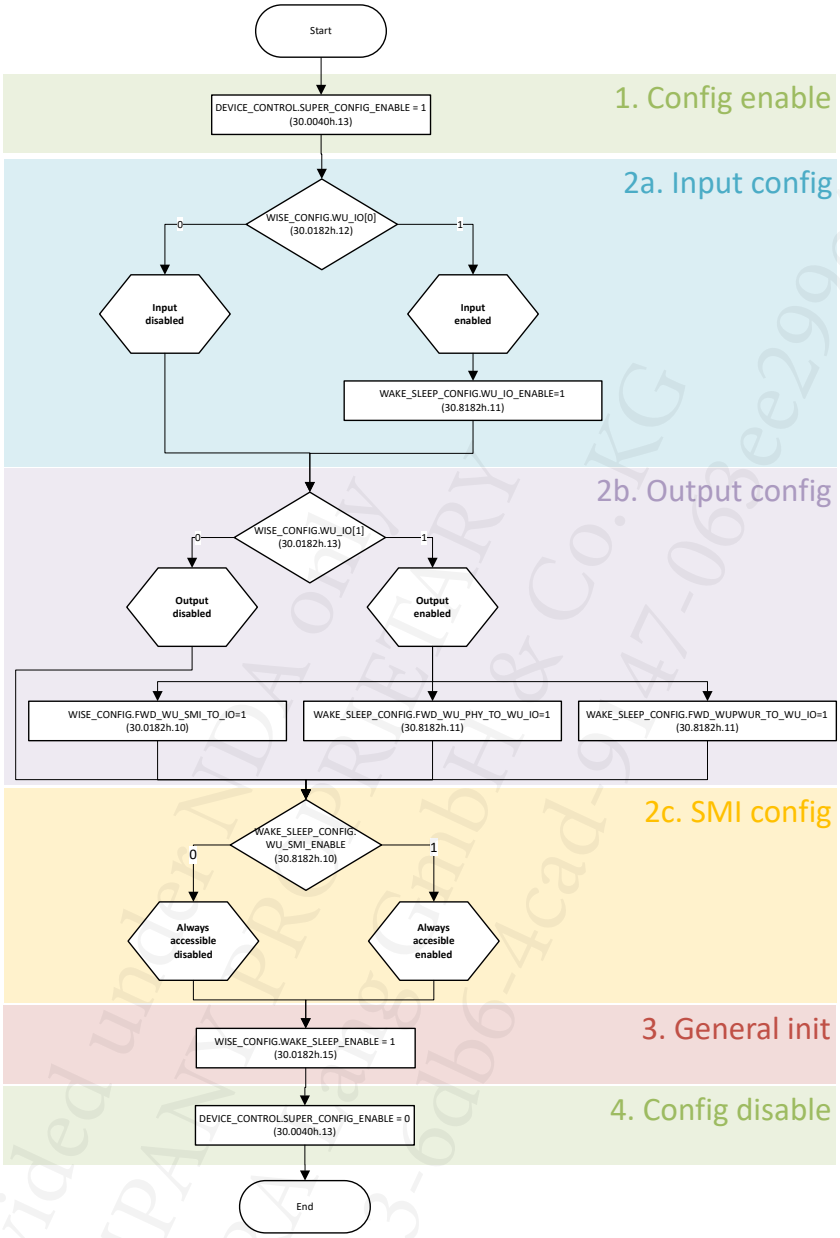


Figure 40. TC10 Configuration

1. Should be enabled only when INH_OVERRIDE (30.0180h.0) = 1. If INH goes low this register can't be accessed.

Not shown in the diagram is the basic wake-sleep mode which allows manual sleep control and local wake-up but no remote wake-up or wake forwarding. For a full set of additional configuration options please have a look at the registers WAKE_SLEEP_CONFIG / WAKE_SLEEP_PARAMETERS / WISE_CONFIG in the TJA1104 datasheet.

Note: *It is possible to put the device into deep sleep without enabling any wakeup source. When entering deep sleep make sure to configure at least one of the following wakeup sources:*

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- *Wake via SMI (WU_SMI_ENABLE must be configured, only available if VDDIO present)*
- *Remote Wake via MDI (FUNCTION_SELECT must be 10 / TC10 Wake-Sleep)*
- *Local Wake via WAKE_IN_OUT (WU_IO_ENABLE must be configured)*

Note: *In case basic wake/sleep is disabled (WISE_CONFIG.WAKE_SLEEP_ENABLE) it is still possible to put the device into deep sleep by transitioning to STANDBY and issuing a sleep request. However as all wakeup functionality is gated by WAKE_SLEEP_ENABLE it is not possible to bring the device out of deep sleep other than through a hardware reset or power-cycle.*

Note: *Some function blocks are active even when WAKE_SLEEP_ENABLE is not set, e.g. WUP_RECEIVED may be asserted. This has no influence on any behavior as all actions are gated by WAKE_SLEEP_ENABLE or WAKE_SLEEP_CONFIG.FUNCTION_SELECT, however it should be considered for software development.*

6.1.3 Operation

During wake-up/sleep operation actions can be triggered via several different paths and observed in various status registers. Figure 41 shows the most common actions involved in TC10 operation.

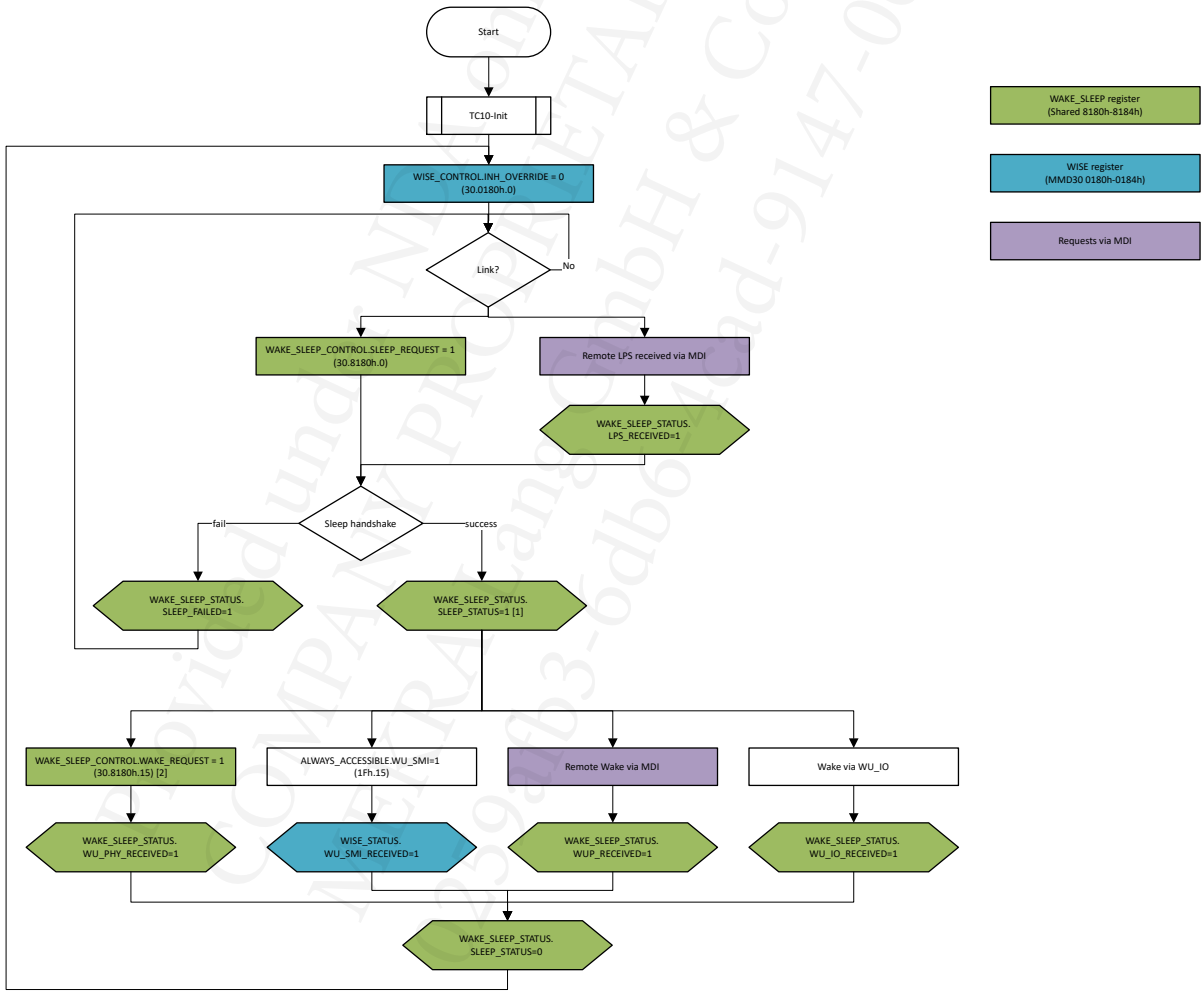


Figure 41. TC10 operation

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1. In case deep sleep was entered WAKE_SLEEP_STATUS cannot be read.
2. Only available if deep sleep was not entered

- Sleep
  - SLEEP_REQUEST via SMI initiates the sleep handshake corresponding to loc_sleep_req in the TC10 specification
  - LPS received via MDI
- Wake-up
  - WAKEUP_PHY_REQUEST via SMI is only possible if the PHY is not in power down. Depending on the configuration and link state a WUP or WUR will be sent via MDI
  - WU_SMI via SMI always accessible register. This register can be accessed even in power down. Only requirement is SMI access which implies VDDIO avaiability. Depending on the configuration the PHY will assert INH and trigger the same actions as for WAKEUP_PHY_REQUEST
  - Wake pulse via WAKE_IN_OUT pin
  - WUP via MDI

INH_OVERRIDE is automatically set after every device reset and wakeup procedure and will keep INHIBIT asserted irrespective of the device state. In order to release INHIBIT when the device entered standby the INH_OVERRIDE bit must be cleared after every reset/wakeup.[8]

6.2 Standby

Standby mode is an intermediate low-power state with PHY receive/transmit functions under reset with the PLL forced into low-power mode. In Standby mode, the PHY preserves its configuration. It is recommended that Device(PHY) reconfiguration is done in this mode.

From an operation point of view, Standby mode corresponds to the IEEE802.3 Power-down mode .

A PHY can enter Standby mode in the following situations.

PHY Device supports Standby mode for:

1. EPHY:

//Enter STANDBY

PHY_CONTROL.GOTO_STANDY (30. 8100. 1) = 1

//Exit STANDBY

PHY_CONTROL.START_OPERATION (30. 8100. 0) = 1

2. SGMII:

//Enter STANDBY

SGMII_ADVANCED_CONTROL.GOTO_STANBDY (30. B010. 1) = 1

//Exit STANDBY

SGMII_ADVANCED_CONTROL.START_OPERATION (30. B010. 0) = 1

3. Complete PHY Device

//Enter STANDBY

DEVICE_CONTROL.GOTO_STANDBY (30.0040h.1) =1

//Exit STANDBY

8 This step is optional. If INH_OVERRIDE is not cleared, wake-up/sleep remains functional with the exception of power down.

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DEVICE_CONTROL.START_OPERATION (30.0040h.0) =1

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7 Compliance Tests

Several externally conducted tests guarantee interoperability between available products as well as compliance with various EMC guidelines. The following sections describe which settings influence the behavior of the TJA1104 towards these compliance tests.

7.1 Impact on compliance

For the interoperability test only the properties of the MDI are relevant. As there are no settings influencing the MDI, neither in a functional nor in a electrical/analog way, there are no specific register values required.

Polarity correction in slave mode is configured via pinstrapping but can be enabled or disabled via register settings as well. For details refer to Section 4.7. In master mode there is no polarity correction.

TC10 Wake/sleep is disabled by default. If it is needed to ensure that TC10 is disabled it can be checked by observing MMD30.WISE_CONFIG.WAKE_SLEEP_ENABLE (30.0182h.15) which acts a global enable switch for all TC10 functionality. For details refer to Section 6.1

7.2 EMC behavior

The EMC behavior is influenced by many factors. One contributor is the type of xMII interface. For that reason all EMC relevant tests were conducted for all available xMII interface (MII, RMII, RGMII and SGMII). Other factors can be influenced via register settings. Section 7.2 shows the specific settings used during EMC testing. Table 41 lists registers as well as other factors potentially changing the EMC behavior. Deviations from the described configuration should be tested carefully on module level EMC tests.

Table 41. EMC relevant registers

No.	Register	Function
1	MMD30.XMII_DATA_IO_CONFIG	Control IO driver strength
2	MMD30.XMII_CLK_IO_CONFIG	Control IO driver strength
3	MMD30.SMI_DATA_IO_CONFIG	Control IO driver strength
4	MMD30.GPIO_IO_CONFIG	Control IO driver strength
5	Shared.PHY_COMPLIANCE_TEST	Output TCLK on IO
6	MMD30.COMPL_TEST_INTF	Output TCLK on IO
7	MMD30.GPIOx_FUNC_CONFIG[1]	Control GPIO functions
8	MMD30.MII_BASIC_CONFIG	Control xMII mode
9	MMD30.XMII_CLK_CONFIG	Control xMII clock settings
10	MMD30.XMII_DATA_CONFIG	Control xMII data settings
11	MMD30.RGMII_TXC_DELAY_CONFIG	Control RGMII txc delay
12	MMD30.RGMII_RXC_DELAY_CONFIG	Control RGMII rxc delay
13	MMD30.SGMII_SR_MII_CTRL	Control SGMII settings

[1] Register present 12 times (GPIO0-GPIO11), not all GPIOs are available on all device variants.

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7.2.1 Configuration

The following table lists the configuration used for all test variants of the IEC 62228-5 test suite. Registers that are not listed have not been changed and contain the power on default value as listed in the datasheet.

Table 42. IEC 62228-5 configuration

Register	Value	Description
Common initialization		
MMD30.DEVICE_CONTROL .SUPER_CONFIG_ENABLE 30.0040h.13	1	Allow configuration changes
MMD1.BASE_T1_PMA_CONTROL .MASTER_SLAVE 01.0834h.14	1/0	Set device to master/slave[1]
MMD30.MII_BASIC_CONFIG .XMII_MODE 30.AFC6h.2:0	4h/5h/7h	Set device to MII/RMII/RGMII[2] **XMII_MODE malfunction:** Writing a non-supported option for MII_BASIC_CONFIG[XMII_MODE] (address 30.AFC6h.2:0) causes an XMII interface malfunction. *Workaround*: Only supported modes should be programmed. Reserved values must be avoided.
Shared.MSE .ENABLE 30.8324h.15	1	Enable MSE calculation
TC10 Sleep (Unwanted wakeup test)		
MMD30.WISE_CONTROL .INH_OVERRIDE 30.0180h.0	0	Release INH as soon as sleep state is reached
MMD30.WISE_CONFIG .WAKE_SLEEP_ENABLE 30.0182h.15	1	Enable global Wake-sleep functionality
Shared.WAKE_SLEEP_CONTROL .SLEEP_REQUEST 30.8180h.0	1	Trigger sleep request[3]
TC10 Wake (Wanted wakeup test)		
MMD30.WISE_CONTROL .INH_OVERRIDE 30.0180h.0	0	Release INH as soon as sleep state is reached
MMD30.WISE_CONFIG .WAKE_SLEEP_ENABLE 30.0182h.15	1	Enable global Wake-sleep functionality
MMD30.WISE_CONFIG .WU_IO 30.0182h.13:12	3	Set WAKE_IN_OUT to input+output(default)
Shared.WAKE_SLEEP_CONFIG .WU_IO_ENABLE 30.8182h.11	1	Enable wakeup on WAKE_IN_OUT pulse
Shared.WAKE_SLEEP_CONTROL .SLEEP_REQUEST 30.8180h.0	1	Trigger sleep request[3]
ALWAYS_ACCESSIBLE .WAKE_REQUEST 1Fh.15	1	Trigger wake request[3]
Powered ESD (Using wanted wakeup configuration)		
MMD30.WISE_CONTROL .INH_OVERRIDE 30.0180h.0	0	Release INH as soon as sleep state is reached
MMD30.WISE_CONFIG .WAKE_SLEEP_ENABLE	1	Enable global Wake-sleep functionality

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Table 42. IEC 62228-5 configuration*...continued*

Register	Value	Description
30.0182h.15		
MMD30.WISE_CONFIG .WU_IO 30.0182h.13:12	3	Set WAKE_IN_OUT to input+output(default)
Shared.WAKE_SLEEP_CONFIG .WU_IO_ENABLE 30.8182h.11	1	Enable wakeup on WAKE_IN_OUT pulse
Shared.WAKE_SLEEP_CONTROL .SLEEP_REQUEST 30.8180h.0	1	Trigger sleep request[3]
ALWAYS_ACCESSIBLE .WAKE_REQUEST 1Fh.15	1	Trigger wake request[3]

[1] DUT set to master, link partner node set to slave
[2] All variants tested during normal mode tests. Fixed to 4h (MII) for all TC10 tests.
[3] Only triggered on DUT, link partner follows automatically.

8 Migration to gigabit Ethernet

The upcoming TJA1120 1000BASE-T1 PHY belongs to the same drop in compatible product family as the TJA1104 which allows an easy migration from TJA1104 to TJA1120. Similar to the TJA1104 which adds MACsec support to TJA1103 there will be a MACsec enabled version of the TJA1120. When following a few rules this is possible even without changing the PCB. The next sections highlight key differences that need attention. All parts that are not covered in these sections can be considered identical between TJA1104 and TJA1120. Of course for detailed information the respective parts of the datasheets should be compared.

8.1 xMII interface

While TJA1104A supports RGMII as well as RMII and MII, TJA1120A only supports RGMII. In order to minimize changes when migrating from TJA1104A to TJA1120A tt is therefore recommended to use RGMII between TJA1104A and MAC. Basic electrical specifications of RGMII are the same for both 100 Mbit/s and 1000 Mbit/s speed grades. These are:

- 6 rx pins (signal direction from PHY to MAC)
- 6 tx pins (signal direction from MAC to PHY)
- I/O voltage (1.8 V, 2.5 V, 3.3 V selectable on PHY)
- 50 ohms impedance

However, keep in mind that the clocks (RXC and TXC) for 100 Mbit/s is 25 MHz while for 1000 Mbit/s it is 125 MHz. This change might impact EMC behavior.

8.2 MDI interface

There are no PCB changes neccessary on the MDI part when migrating between TJA1104A and TJA1120A as the few components required on the MDI interface are either the same or at least footprint compatible between 100BASE-T1 and 1000BASE-T1 application:

- Common mode choke (CMC)
- AC coupling capacitors
- DC network
- Requirement dependant: ESD diodes

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Except for the CMC, all components can be choosen to fit 100 Mbit/s and 1000 Mbit/s use cases while CMCs for those use cases share the same footprint allowing the migration with a simple assembly option.

The connector choosen for the MDI interface is dependant on the application so no general recommendation can be made. Current experience suggests that 100BASE-T1 can safely be used on unshielded twisted pair (UTP) cables while 1000BASE-T1 requires shielding (STP). There are solutions available that offer both options in the same connector family, again avoiding PCB changes. So in case connector compatibility is required such a solution is recommended.

8.3 Supply

The supply configuration of the upcoming TJA1120 1000BASE-T1 PHY was chosen to match as much as possible a TJA1104 supply configuration. The following table highlights specific parts of the application that should be considered to keep efforts in the migration process as small as possible.

Table 43. TJA1120 vs TJA1104 differences

Section	Recommendation	Details
Digital supply (VDD_CORE)	Supply via external 1.1V	Both variants support external 1.1V (individual supply mode)
Digital supply (VREGD_IN/ VDD_CORE)[1]	Supply via external 1.1V	
Analog supply (VREGA_IN)	Supply via 2.5V/3.3V	Both variants support 2.5V/3.3V input at this pin
Analog supply (TJA1104: VREGA_OUT/ VDDA_TRX) (TJA1120: VDDA_RX/VDDA_ TRX)	Prepare 2 assembly options: TJA1120: Supply via 1.8V TJA1104: Disconnect ext. supply	TJA1104 provides 2.5V on VREGA_OUT for VDDA_TRX TJA1120 requires external 1.8V supply
Digital interface I/O voltage	Both devices support 1.8V, 2.5V, 3.3V VDDIO level	1.8V recommended as it is expected to yield best performance for 1000Mbit/s RGMII

[1] Pin 4 was renamed from VREGD_IN to VDD_CORE for TJA1120

The following schematic shows the option described above:

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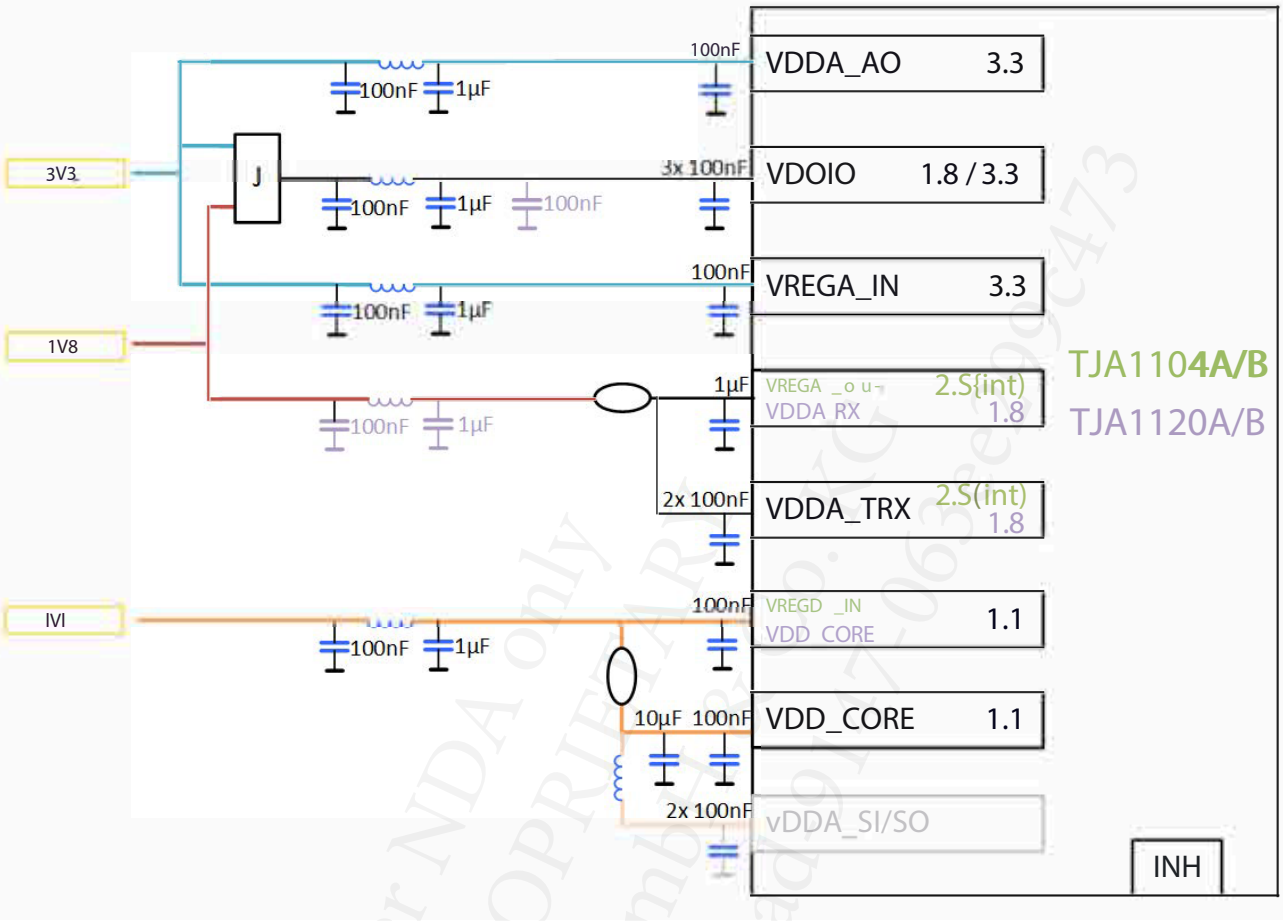


Figure 42. TJA1104 to TJA1120 migration supply part

8.4 Registers

Registers are kept identical or compatible where possible.

Real-time drivers (RTD) will be provided supporting both TJA1104 and TJA1120 allowing effortless migration on software level as well.

8.5 Miscellaneous

The oscillator used on TJA1120 offers LVDS as clock input as additional mode as well as an improved single ended mode compared single ended mode and crystal mode supported by TJA1104. For minimum migration effort it is recommended to use one of the modes supported by TJA1104 also for TJA1120.

The xMII group of the pinstrapping configuration changed. Modes other than RGMII are not available for TJA1120. In addition some RGMII modes have been reorganized. Consult the respective section in the datasheet to find the correct configuration.

The miscellaneous group of the pinstrapping configuration changed. Polarity correction is mandatory in the 1000BASE-T1 standard and therefore not configurable anymore. Instead the oscillator mode previously mentioned needs to be configured via pinstrapping

Note: *Be careful when setting oscillator mode (CLK select) as wrong configurations may damage the device.*

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9 PIN_FMEA

Table 44. PIN_FMEA

Failure classification

A – Damage to device / Bus may be affected
B – No damage to device / No bus communication possible
C – No damage to device / Bus communication possible / Corrupted node excluded from communication
D – No damage to device / Bus communication possible / Reduced functionality of device
n/a – Not applicable

		Short to neighbor				Short to GND		Pin open		Short to VDDIO		Short to 1.1 V		Short to VDDAO	
Pin	Pin name	Class	Pin	Pin name	Mode	Class	Mode	Class	Mode	Class	Mode	Class	Mode	Class	Mode
TJA1103A															
1	MDC	D	2	INT_N	Loss of IRQ and MDIO functionality	D	Loss of IRQ and MDIO functionality	D	Loss of IRQ and MDIO functionality	D	Loss of IRQ and MDIO functionality	D	Loss of IRQ and MDIO functionality	D	Loss of IRQ and MDIO functionality
2	INT_N	A	3	VDD_CORE	Limiting value of VDD_CORE exceeded	D	Loss of IRQ functionality IRQ constantly asserted	D	Loss of IRQ functionality	D	Loss of IRQ functionality IRQ constantly deasserted	D	Loss of IRQ functionality	D	Loss of IRQ functionality IRQ constantly deasserted Potential damage to SoC depending on limiting values on IRQ pin
3	VDD_CORE	A	4	VREGD_IN	Limiting value of VDD_CORE exceeded	B	Device in undervoltage	B	Device in undervoltage	A	Limiting value of VDD_CORE exceeded	n/a		A	Limiting value of VDD_CORE exceeded
4	VREGD_IN	A	5	XO	Limiting value of XO exceeded	B	Device in undervoltage	B	Device in undervoltage	B	Device in undervoltage if VDDIO is not sufficiently high Potential damage of IO pins due to increased VDDIO	B	Device in undervoltage	D	Reduced EMC /supply filtering performance
5	XO	B	6	XI / CLK_IN	No digital clock present	B	No digital clock available, no communication possible	B	No digital clock available, no communication possible	A	Limiting value of XO exceeded	B	No digital clock available, no communication possible	A	Limiting value of XO exceeded
6	XI / CLK_IN	A	7	VREGA_IN	Limiting value of XI exceeded	B	No digital clock available, no communication possible	B	No digital clock available, no communication possible	A	Limiting value of XO exceeded	B	No digital clock available, no communication possible	A	Limiting value of XO exceeded
7	VREGA_IN	A	8	VREGA_OUT	Limiting value of analog transmitter supply exceeded	B	AFE not powered No communication possible	B	AFE not powered No communication possible	B	If VDDIO is not sufficiently high, AFE not powered No communication possible	B	AFE not powered No communication possible	D	Reduced EMC /supply filtering performance
8	VREGA_OUT	A	9	VDDA_AO	Limiting value of analog transmitter supply exceeded	B	AFE not powered No communication possible	B	AFE not powered No communication possible	B	If VDDIO is not sufficiently high, AFE not powered No communication possible	B	AFE not powered No communication possible	A	Limiting value of analog transmitter supply exceeded
9	VDDA_AO	D	10	RST_N	Loss of RST_N control via SoC Internal POR and SW remains functional	B	Always on domain unpowered No communication possible	B	Always on domain unpowered No communication possible	B	Depends on level and state of VDDIO For VDDIO lower than 3.3 V, the always on domain is in undervoltage	B	Always on domain is in undervoltage	n/a	

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Table 44. PIN_FMEA*...continued*

Failure classification

A – Damage to device / Bus may be affected
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C – No damage to device / Bus communication possible / Corrupted node excluded from communication
D – No damage to device / Bus communication possible / Reduced functionality of device
n/a – Not applicable

		Short to neighbor				Short to GND		Pin open		Short to VDDIO		Short to 1.1 V		Short to VDDAO	
Pin	Pin name	Class	Pin	Pin name	Mode	Class	Mode	Class	Mode	Class	Mode	Class	Mode	Class	Mode
10	RST_N	B	11	WAKE_IN_OUT	Device will be kept in reset due to WAKE_IN_OUT pulldown	B	Device is reset	D	Loss of RST_N control via SoC Internal POR and SW remains functional	B	Depends on level and state of VDDIO Device will be kept in reset for sufficiently low VDDIO levels	B	Spurious device reset	D	Loss of RST_N pin function SW reset still operational
11	WAKE_IN_OUT	B	12	INH	If INH is used and connected to SBC/SMPS, potential loss of voltage supply No effect, if INH and wake/sleep is not used in application	D	If wake/sleep not used, no effect Loss of wake_in_out mechanism	D	If wake/sleep not used, no effect Loss of wake_in_out mechanism and spurios wakeups if internal pull-down is not enabled	D	If wake/sleep not used, no effect Possible corruption of SoC IO pins (VDDIO domain) via INH (always-on domain) Spurious wake/wake_forwarding events	D	If wake/sleep not used, no effect Spurious wake/wake_forwarding events	D	If wake/sleep not used, no effect Spurious wake/wake_forwarding events
12	INH	A	13	VDDA_TRX	Limiting value of analog transmitter supply exceeded	D	No effect, if wake/sleep not used Early SMPS/SBC shutdown and loss of supply	D	No effect, if wake/sleep not used Early SMPS/SBC shutdown and loss of supply	D	No effect, if wake/sleep not used Potential loss of voltage supply if INH is connected to SBC/SMPS	D	No effect, if wake/sleep not used Potential loss of voltage supply if INH is connected to SBC/SMPS	D	No effect if wake/sleep not used Loss of INH shut-down functionality
13	VDDA_TRX	B	14	TRX_P	Loss of communication	A	Loss of communication Device may get damaged	B	No communication	A	No communication Limiting value of analog transmitter may get exceeded	B	No communication	A	No communication Limiting value of analog transmitter may get exceeded
14	TRX_P	B	15	TRX_M	Loss of communication	B	Loss of communication	B	Loss of communication	B	Loss of communication	B	Loss of communication	B	Loss of communication
15	TRX_M	B	16	VDDA_TRX	Loss of communication	B	Loss of communication	B	Loss of communication	B	Loss of communication	B	Loss of communication	B	Loss of communication
16	VDDA_TRX	B	17	GPIO0 / TMS / CONFIG0	Loss of communication	A	Loss of communication Device may get damaged	B	No communication	A	No communication Limiting value of analog transmitter may get exceeded	B	No communication	A	No communication Limiting value of analog transmitter may get exceeded
17	GPIO0 / TMS / CONFIG0	D	18	GPIO1 / TDI	Loss of GPIO0/1 function, Loss of JTAG function, Wrong PHY address latching No MDIO access / potentially conflicting PHY address	D	Loss of GPIO0 function, Loss of JTAG function, Wrong PHY address latching No MDIO access / potentially conflicting PHY address	D	Loss of GPIO0 function, Loss of JTAG function, Wrong PHY address latching No MDIO access / potentially conflicting PHY address	D	Loss of GPIO0 function, Loss of JTAG function, Wrong PHY address latching No MDIO access / potentially conflicting PHY address	D	Loss of GPIO0 function, Loss of JTAG function, Wrong PHY address latching No MDIO access / potentially conflicting PHY address	D	Loss of GPIO0 function, Loss of JTAG function, Wrong PHY address latching No MDIO access / potentially conflicting PHY address
18	GPIO1 / TDI	D	19	RX_ER / GPIO2 / TDO / CONFIG1	Loss of GPIO0/1 function, Loss of JTAG function Communication possible if RX_ER	D	Loss of JTAG / GPIO1 function	D	Loss of JTAG / GPIO1 function	D	Loss of JTAG / GPIO1 function	D	Loss of JTAG / GPIO1 function	D	Loss of JTAG / GPIO1 function

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Table 44. PIN_FMEA*...continued*

Failure classification

A – Damage to device / Bus may be affected
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D – No damage to device / Bus communication possible / Reduced functionality of device
n/a – Not applicable

		Short to neighbor				Short to GND		Pin open		Short to VDDIO		Short to 1.1 V		Short to VDDAO	
Pin	Pin name	Class	Pin	Pin name	Mode	Class	Mode	Class	Mode	Class	Mode	Class	Mode	Class	Mode
					not used in applcation.										
19	RX_ER / GPIO2 / TDO / CONFIG1	B	20	RXD3 / GPIO3 / CONFIG2	Loss of JTAG Erroneous pinstrap configuration read Erroneous RX communication Loss of GPIO functionality	D	Loss of JTAG Erroneous pinstrap configuration read Erroneous RX communication Loss of GPIO functionality	D	Loss of JTAG Erroneous pinstrap configuration read Erroneous RX communication Loss of GPIO functionality	D	Loss of JTAG Erroneous pinstrap configuration read Erroneous RX communication Loss of GPIO functionality	D	Loss of JTAG Erroneous pinstrap configuration read Erroneous RX communication Loss of GPIO functionality	D	Loss of JTAG Erroneous pinstrap configuration read Erroneous RX communication Loss of GPIO functionality
20	RXD3 / GPIO3 / CONFIG2	B	21	RXD2 / GPIO4 / CONFIG3	Erroneous pinstrap configuration read Erroneous RX communication in MII/RGMII mode Loss of GPIO functionality	B	Erroneous pinstrap configuration read Erroneous RX communication in MII/RGMII mode Loss of GPIO functionality	B	Erroneous pinstrap configuration read Erroneous RX communication in MII/RGMII mode Loss of GPIO functionality	B	Erroneous pinstrap configuration read Erroneous RX communication in MII/RGMII mode Loss of GPIO functionality	B	Erroneous pinstrap configuration read Erroneous RX communication in MII/RGMII mode Loss of GPIO functionality	B	Erroneous pinstrap configuration read Erroneous RX communication in MII/RGMII mode Loss of GPIO functionality
21	RXD2 / GPIO4 / CONFIG3	B	22	VDDIO	Erroneous pinstrap configuration read Erroneous RX communication in MII/RGMII mode Loss of GPIO functionality	B	Erroneous pinstrap configuration read Erroneous RX communication in MII/RGMII mode	B	Erroneous pinstrap configuration read Erroneous RX communication in MII/RGMII mode	B	Erroneous pinstrap configuration read Erroneous RX communication in MII/RGMII mode	B	Erroneous pinstrap configuration read Erroneous RX communication in MII/RGMII mode	B	Erroneous pinstrap configuration read Erroneous RX communication in MII/RGMII mode
22	VDDIO	B	23	RXD1 / CONFIG4	Erroneous pinstrap configuration read Erroneous RX communication in MII/RGMII mode	B	Loss of GPIOs, MII/ RMII/RGMII, SMI and IRQ access	B	Erroneous pinstrap configuration read Erroneous RX communication in MII/RGMII mode	n/a		B	Loss of GPIOs, MII/ RMII/RGMII, SMI and IRQ access Potential corruption of Ics on 1.1 V domain due to increased voltage (VDDIO)	B	Loss of GPIOS, MII/ RMII/RGMII, SMI and IRQ access Potential corruption of Ics on 1.1 V domain due to increased voltage (VDDIO)
23	RXD1 / CONFIG4	B	24	RXD0 / CONFIG5	Erroneous pinstrap configuration read Erroneous RX communication	B	Erroneous pinstrap configuration read Erroneous RX communication	B	Erroneous pinstrap configuration read Erroneous RX communication	B	Erroneous pinstrap configuration read Erroneous RX communication	B	Erroneous pinstrap configuration read Erroneous RX communication	B	Erroneous pinstrap configuration read Erroneous RX communication
24	RXD0 / CONFIG5	B	25	RX_CTL / CONFIG6	Erroneous pinstrap configuration read Erroneous RX communication	B	Erroneous pinstrap configuration read Erroneous RX communication	B	Erroneous pinstrap configuration read Erroneous RX communication	B	Erroneous pinstrap configuration read Erroneous RX communication	B	Erroneous pinstrap configuration read Erroneous RX communication	B	Erroneous pinstrap configuration read Erroneous RX communication
25	RX_CTL / CONFIG6	B	26	RXC / GPIO8 / TX_TCLK	Erroneous pinstrap configuration read Erroneous RX communication	B	Erroneous pinstrap configuration read Erroneous RX communication	B	Erroneous pinstrap configuration read Erroneous RX communication	B	Erroneous pinstrap configuration read Erroneous RX communication	B	Erroneous pinstrap configuration read Erroneous RX communication	B	Erroneous pinstrap configuration read Erroneous RX communication

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Table 44. PIN_FMEA*...continued*

Failure classification

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D – No damage to device / Bus communication possible / Reduced functionality of device
n/a – Not applicable

		Short to neighbor				Short to GND		Pin open		Short to VDDIO		Short to 1.1 V		Short to VDDAO	
Pin	Pin name	Class	Pin	Pin name	Mode	Class	Mode	Class	Mode	Class	Mode	Class	Mode	Class	Mode
26	RXC / GPIO8 / TX_TCLK	B	27	VDDIO	Erroneous pinstrap configuration read Erroneous RX communication or loss of GPIO8 functionality (if used)	B	Erroneous pinstrap configuration read Erroneous RX communication or loss of GPIO8 functionality (if used)	B	Erroneous pinstrap configuration read Erroneous RX communication or loss of GPIO8 functionality (if used)	B	Erroneous pinstrap configuration read Erroneous RX communication or loss of GPIO8 functionality (if used)	B	Erroneous pinstrap configuration read Erroneous RX communication or loss of GPIO8 functionality (if used)	B	Erroneous pinstrap configuration read Erroneous RX communication or loss of GPIO8 functionality (if used)
27	VDDIO	D	28	TX_ER / GPIO9 / TCK	Loss of JTAG function Loss of GPIO function Loss of TX_ER function, TX communication error if TX_ER pin is used in application	B	Loss of GPIOs, MII/ RMII/RGMII, SMI and IRQ access	B	Loss of GPIOs, MII/ RMII/RGMII, SMI and IRQ access	n/a		B	Loss of GPIOs, MII/ RMII/RGMII, SMI and IRQ access Potential corruption of Ics on 1.1 V domain due to increased voltage (VDDIO)	B	Loss of GPIOs, MII/ RMII/RGMII, SMI and IRQ access Potential corruption of Ics on 1.1 V domain due to increased voltage (VDDIO)
28	TX_ER / GPIO9 / TCK	B	29	TXC	Loss of GPIO9 function Loss of JTAG function Loss of TX communication	B	Loss of GPIO9 function Loss of JTAG function Loss of TX communication if TX_ER is used in application	B	Loss of GPIO9 function Loss of JTAG function Loss of TX communication if TX_ER is used in application	B	Loss of GPIO9 function Loss of JTAG function Loss of TX communication if TX_ER is used in application	B	Loss of GPIO9 function Loss of JTAG function Loss of TX communication if TX_ER is used in application	B	Loss of GPIO9 function Loss of JTAG function Loss of TX communication if TX_ER is used in application
29	TXC	B	30	TX_CTL	Loss of TX communication	B	Loss of TX communication	B	Loss of TX communication	B	Loss of TX communication	B	Loss of TX communication	B	Loss of TX communication
30	TX_CTL	B	31	TXD0	Loss of TX communication	B	Loss of TX communication	B	Loss of TX communication	B	Loss of TX communication	B	Loss of TX communication	B	Loss of TX communication
31	TXD0	B	32	TXD1	Loss of TX communication in MII/RGMII mode Loss of GPIO10 and GPIO 11 functions	B	Loss of TX communication in MII/RGMII mode Loss of GPIO10 function	B	Loss of TX communication in MII/RGMII mode Loss of GPIO10 function	B	Loss of TX communication in MII/RGMII mode Loss of GPIO10 function	B	Loss of TX communication in MII/RGMII mode Loss of GPIO10 function	B	Loss of TX communication in MII/RGMII mode Loss of GPIO10 function
32	TXD1	B	33	TXD2 / GPIO10	Loss of TX communication mode Loss of GPIO10 functions	B	Loss of TX communication in MII/RGMII mode Loss of GPIO11 function	B	Loss of TX communication in MII/RGMII mode Loss of GPIO11 function	B	Loss of TX communication in MII/RGMII mode Loss of GPIO11 function	B	Loss of TX communication in MII/RGMII mode Loss of GPIO11 function	B	Loss of TX communication in MII/RGMII mode Loss of GPIO11 function
33	TXD2 / GPIO10	B	34	TXD3 / GPIO11	Loss of TX communication mode (MII/ RGMII mode) Loss of GPIOs	B	Loss of TX communication mode (MII/ RGMII mode) Loss of GPIOs	B	Loss of TX communication mode (MII/ RGMII mode) Loss of GPIOs	B	Loss of TX communication mode (MII/ RGMII mode) Loss of GPIOs	B	Loss of TX communication mode (MII/ RGMII mode) Loss of GPIOs	B	Loss of TX communication mode (MII/ RGMII mode) Loss of GPIOs
34	TXD3 / GPIO11	B	35	VDDIO	Loss of TX communication mode (MII/ RGMII mode) Loss of GPIOs	B	Loss of TX communication mode (MII/ RGMII mode) Loss of GPIOs	B	Loss of TX communication mode (MII/ RGMII mode) Loss of GPIOs	B	Loss of TX communication mode (MII/ RGMII mode) Loss of GPIOs	B	Loss of TX communication mode (MII/ RGMII mode) Loss of GPIOs	B	Loss of TX communication mode (MII/ RGMII mode) Loss of GPIOs
35	VDDIO	D	36	MDIO	Loss of MDIO function	B	Loss of GPIOs, MII/ RMII/RGMII,	B	Loss of GPIOs, MII/ RMII/RGMII,	n/a		B	Loss of RX/TX communication Potential corruption of	B	Loss of RX/RX communication

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Table 44. PIN_FMEA*...continued*

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D – No damage to device / Bus communication possible / Reduced functionality of device
n/a – Not applicable

		Short to neighbor				Short to GND		Pin open		Short to VDDIO		Short to 1.1 V		Short to VDDAO	
Pin	**Pin name**	**Class**	**Pin**	**Pin name**	**Mode**	**Class**	**Mode**	**Class**	**Mode**	**Class**	**Mode**	**Class**	**Mode**	**Class**	**Mode**
							SMI and IRQ access		SMI and IRQ access				Ics on 1.1 V domain due to increased voltage (VDDIO)		Potential corruption of ICs on VDDIO domain due to increased voltage (VDDAO)
36	MDIO	D	1	MDC	Loss of MDIO function	D	Loss of MDIO function	D	Loss of MDIO function	D	Loss of MDIO function	D	Loss of MDIO function	D	Loss of MDIO function
37	GNDPAD														
TJA1103B															
28	GPIO9 / TCK	B	29	VDDA_SO	Loss of JTAG function Loss of GPIO9 function Loss of SGMII connectivity	D	Loss of JTAG and GPIO9 function	D	Loss of JTAG and GPIO9 function	D	Loss of JTAG and GPIO9 function	D	Loss of JTAG and GPIO9 function	D	Loss of JTAG and GPIO9 function
29	VDDA_SO	A	30	SON	SGMII PMA may get damaged	B	Loss of SGMII connectivity	B	Loss of SGMII connectivity	A	Limiting value of SGMII supply exceed	n/a		A	Limiting value of SGMII supply exceed
30	SON	B	31	SOP	Loss of RX communication	B	Loss of RX connectivity	B	Loss of RX connectivity	A	Loss of RX connectivity, SGMII PMA may get damaged	A	Loss of RX connectivity, SGMII PMA may get damaged	A	Loss of RX connectivity, SGMII PMA may get damaged
31	SOP	B	32	SIP	Loss of RX/TX communication	B	Loss of RX connectivity	B	Loss of RX connectivity	A	Loss of RX connectivity, SGMII PMA may get damaged	A	Loss of RX connectivity, SGMII PMA may get damaged	A	Loss of RX connectivity, SGMII PMA may get damaged
32	SIP	B	33	SIN	Loss of TX communication	B	Loss of TX communication	B	Loss of TX communication	A	Loss of TX connectivity, SGMII PMA may get damaged	A	Loss of TX connectivity, SGMII PMA may get damaged	A	Loss of TX connectivity, SGMII PMA may get damaged
33	SIN	A	34	VDDA_SI	SGMII PMA may get damaged	B	Loss of TX communication	B	Loss of TX communication	A	Loss of TX connectivity, SGMII PMA may get damaged	A	Loss of TX connectivity, SGMII PMA may get damaged	A	Loss of TX connectivity, SGMII PMA may get damaged
34	VDDA_SI	A	35	VDDIO	SGMII PMA may get damaged	B	Loss of SGMII connectivity	B	Loss of SGMII connectivity	A	Limiting value of SGMII supply exceed	n/a		A	Limiting value of SGMII supply exceed
35	VDDIO	D	1	MDC	Loss of MDIO function	B	Loss of SMI and IRQ access	B	Loss of RX communication	n/a		B	Loss of RX communication Potential corruption of Ics on 1.1 V domain due to increased voltage (VDDIO)	B	Loss of RX communication Potential corruption of Ics on 1.1 V domain due to increased voltage (VDDIO)

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10 PCB Checklist

Table 45. PCB Checklist

Pin name	Pin number	TJA1104 Requirement
		Check pinning and correct pin numbers.
MDC	1	MDC pin must be connected to the MDC output of the MDIO master device (typically to the µC). Provision optional series resistor at the driver (close the uC) to improve EMC performance. The impedance of the series resistor plus the output impedance of the MDC driver shall match the impedance of the PCB trace.
		If MDC is distributed to multiple PHYs, keep the branches and stubs as short as possible.
INT_N	2	If IRQ_N is used, connect pull-up resistor of (e.g. 10 kΩ) to VDDIO. This can be integrated pull-up of the µC. The value of the pull-up resistor should be chosen such that a proper high/low level is ensured.
		If not used it can be left open.
VDD_CORE	3	Connect a 100 nF and a 10 µF ceramic capacitor when using single supply variant.
		Connect to VREGD_IN and only a 100 nF capacitor when using individual supply variant. VREGD_IN needs to be 1.1 V in this case.
VREGD_IN	4	Connect a 100 nF and a 1 µF ceramic capacitor between this pin and GND close to the pin. Add a ferrite bead for filtering. See Section 3.5 for details.
XO	5	If this pin is not used, it can be left open.
		If a 25 MHz external crystal (+-100ppm) is used, follow the crystal supplier recommendations. Provision for optional series resistor.
XI	6	If this pin is not used, it should be shorted to GND to avoid noise input. Provision for optional series resistor.
VREGA_IN	7	Connect a 100 nF and a 1 µF ceramic capacitor between this pin and GND close to the pin. Add a ferrite bead for filtering. See Section 3.5 for details.
VREGA_OUT	8	Connect a 1 µF ceramic capacitor when using single supply variant. Connect to VREGA_IN and omit additional 1 µF capacitor when using individual supply variant.
		Connect to VREGA_IN and omit additional 1 µF capacitor when using individual supply variant.
VDDA_AO	9	Connect a 100 nF ceramic capacitor between this pin and GND close to the pin.
RST_N	10	Connect to GPIO of the µC. A pull-up resistor of about 10 kΩ to VDD_AO possible, but not necessary.
WAKE_IN_OUT	11	If used as wake-up output, the pin has a high-side switch to VDDA_AO level when activated, otherwise it is hi-Z.
		If not used, connect a pull-down resistor to ensure a proper low level.
VDDA_TRX	13, 16	The two VDDA_TRX pins and the VREG_OUT should be connected together on the PCB.
TRX_P	14	The recommended circuit can be found in Section 3.4.

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Table 45. PCB Checklist*...continued*

Pin name	Pin number	TJA1104 Requirement
TRX_M	15	
RXD3	20	No external termination resistors are necessary. 0 ohm resistors for fine tuning recommended.
RXD2	21	
RXD1	23	
RXD0	24	
RXD0	24	
RX_CTL	25	
RX_RX_ER	19	
RXC	26	(Depending on xMII mode) No external termination resistors are necessary. 0 ohm resistors recommended for fine tuning .
TXC	29	
VDDIO	22, 27, 35	The three VDDIO pins should be connected together on the PCB.
TXD0	31	No external resistors are necessary. 0 ohm resistors recommended for fine tuning
TXD1	32	
TXD2	33	
TXD3	34	
TX_CTL	3	
TX_ER	28	
MDIO	36	Connect to MDIO pin of µC with a pull-up resistor of 1.5 kΩ to 10 kΩ to $V_{DD(IO)}$, the pull-up can be also within the µC. Make sure the pull-up resistor is available only once at the same MDIO bus.
GNDPAD	37	The exposed die pad shall be soldered to GND with a low impedance. It is recommended to have a ring of GND vias for optimized EMC behavior.

11 Abbreviations

Table 46. Abbreviations

Term	Definition
CDM	Charge-device model
CDR	Clock-data recovery
CMC	Common-mode choke
EMC	Electromagnetic compatibility
ESD	Electrostatic discharge
FCS	Frame Check Sequence
HBM	Human-body model
MDI	Media Dependent Interface
MDIO	Management Data Input/Output
MII	Media Independent Interface
PHY	Physical Layer

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Table 46. Abbreviations...*continued*

Term	Definition
RGMII	Reduced Gigabit Media Independent Interface
RMII	Reduced Media Independent Interface
SGMII	Serial Gigabit Media Independent Interface

12 References

[1] TJA1104 data sheet, secure files at nxp.com/TJA1104

[2] AH544111_4 PCB design guidelines, Secure Files at nxp.com/TJA1103

[3] IEEE 802.3bw-2015, Physical Layer Specifications and Management Parameters for 100 Mb/s Operation over a Single Balanced Twisted Pair Cable (100BASE-T1)

[4] IEC 62228-5 2019-09 Integrated curciuts - EMC evaluation of tranceivers - Part 5: Ethernet transceivers

[5] Open Alliance 100BASE-T1 Ethernet Channel & Components (TC2)

13 Revision history

Table 47. Revision history

Document ID	Release date	Description
AN13663 v.1.0	16 April 2024	• Initial version

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